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# Optimizing Startup Software Product Development through Generative AI

## Investigating the value of Active Persona-generated feedback in assessing software quality and user alignment

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This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfilment of the requirements for the degree of Master of Science in Software Engineering. The thesis is equivalent to 20 weeks of full-time studies.

The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

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# ABSTRACT

**Background.** The concept of personas for user simulation, introduced by Cooper in 1999, aimed to improve software design by providing detailed user representations. This addressed the issue of engineers designing for abstract "elastic users," leading to poor usability despite technical functionality. Pruitt and Adlin later refined this by emphasizing data-driven profiles capturing behavioral patterns. However, Pruitt and Grudin, followed by Chapman and Milham, highlighted implementation challenges and the "representational paradox"—the difficulty in proving persona validity. The static nature of traditional personas became problematic with rapid technological advancements, as noted by Salminen et al. Despite drawbacks, personas remained valuable for humanizing design. AI research between 2015 and 2020 explored AI-driven personas for more accurate, data-driven user representations. The rise of generative AI further increased interest in AI's potential to create dynamic user personas, though careful validation is necessary due to potential unreliability.

**Objectives.** This research aims to validate the concept of Active Personas (APs) as an effective tool for generating contextual software development feedback. By systematically analyzing the efficiency of AI-generated Active Personas, the study will thoroughly examine their ability to produce meaningful feedback that matches or exceeds real user feedback.

**Methods.** The research project will follow the Design Science Research (DSR) methodology to create Active Personas as artefacts with the objective of generating contextual feedback on selected codebases. By choosing the DSR framework as the research methodology, the aim is to create practical artefacts that address real-world challenges in the product development cycle.

**Results.** The study demonstrated that Active Personas (APs) powered by generative AI can efficiently implement diverse user perspectives based on personas and generate contextual feedback aligned with them and established usability principles. GPT-4o showed the fastest execution for image-based UI tasks (1m 14s), while LLaMA3 was the most efficient for code analysis (30m 12s). The feedback generated by APs with GPT-4 strongly aligned with usability standards and often matched real user feedback. In cases like readability and character display integrity, APs' insights not only validated user concerns but also provided deeper analysis and improvement suggestions. In OS-specific issues and codebase analysis, APs offered more generic and speculative technical analysis. APs that use a codebase analysis approach tend to be less efficient, context-aligned, and trustworthy.

**Conclusions.** APs proved to be a valuable tool for early-stage usability evaluation, offering efficiency and depth while highlighting certain gaps in real-world alignment. The application of APs in a startup environment can bring value to practitioners in the product development cycle, but human supervision in the evaluation process remains a key factor in ensuring contextual accuracy and mitigating potential AP-generated feedback inaccuracy.

**Keywords: Generative AI, Persona, Feedback, Usability, User-centered design**

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# 1 INTRODUCTION

The active involvement of users and the consideration of their feedback throughout the software development lifecycle are crucial factors in ensuring the success of a software product. Their input is essential for requirements engineering artifacts such as user scenarios, concept maps, and, more importantly, personas. In the research article *"Personas: From Theory to Practices"* by Chang, Lim, and Stoltman, a persona is defined as a precise description of a user's characteristics and what they want to accomplish [1]. First introduced by Cooper in 1999 in his book *The Inmates Are Running the Asylum*, this concept emphasizes that personas are created to help designers understand, describe, and maintain a clear focus on users' goals and behavioral patterns [2]. Personas have played a significant role as a requirement engineering artifact. Their importance has evolved over the years, becoming a fundamental tool in user-centered design, providing a clear picture of users' needs, goals, and behaviors, ensuring that the targeted software meets real user needs. Nowadays, personas improve communication between stakeholders and development teams, guide decision-making, and help prioritize features. They remain a key element in agile, human-centered design and serve as a strong reference point in the development of intuitive and user-friendly software. Despite their importance, obtaining meaningful user participation for feedback collection remains a recurring challenge throughout the software development lifecycle.

Multiple factors hinder effective user involvement in the product development phase, including organizational scheduling constraints, budget limitations, and difficulties in gathering a representative set of the user population [3]. Development teams, especially startups, often struggle with scheduling conflicts when coordinating with external users, as their resources are limited. Besides, many organizations lack established methodologies for systematically collecting and integrating user feedback in their product development cycle [4], resulting in inconsistent results. Meanwhile, internal stakeholders may prioritize business requirements over end-user needs [5]. These limitations regarding user participation in the development process have significant consequences, requirements engineering deficiencies being one of them, originating directly from insufficient user involvement. These deficiencies have been consistently identified as a primary factor in quality issues leading to project failures [6], particularly in cases where products are developed with minimal user input—often resulting in an overload of unnecessary features and the omission of essential functionalities that address real user needs [7]**.**

The mismatch between implemented features and user needs leads to lower adoption rates, poor user satisfaction, and an increase in support costs as end users struggle with incomplete workflows [7]. For startups, these challenges can be very detrimental—they are often the fatal blow that prevents them from reaching the market and cause them to fail shortly after launch. Research indicates that approximately 42% of startups fail due to a lack of market need for their product [8], a high percentage that highlights the critical importance of aligning product development with user needs to ensure long-term success.

This research project explores new approaches to addressing this challenge through Active Personas (APs)—AI-powered fictional users that generate contextual feedback based on active user interface interactions. The primary objective of this research is to validate the concept of Active Personas as a tool to generate contextual feedback and improve the software product development cycle. This project implements modern generative AI technologies, including Meta Llama 3, but also the computer vision-based AI model from OpenAI, among others, to evaluate two carefully selected open-source applications representing different levels of user interface (UI) complexity, in which APs will simulate user interaction and generate meaningful feedback aligned with actual user design principles.

The goal is to assess the potential of APs to support development teams in identifying usability strengths, weaknesses, and issues, and improving overall product–market fit.

The second objective of this research project is to critically evaluate the efficiency of AI-generated Active Personas by systematically analyzing the contextual relevance and alignment of their feedback with authentic user experience paradigms in the software product development cycle. Following these objectives, this thesis will proceed by first reviewing existing works and their relevance related to user participation in product development, the personas concept, and recent applications of generative AI technologies. Subsequently, this research paper introduces a comprehensive explanation of the research methodology selected for this research project, including a detailed description of the design and implementation of the Active Personas and the framework used for evaluating their potential in generating contextual feedback. The results section presents a comprehensive examination of the feedback generated by APs through simulated interactions, emphasizing alignment with user-centered design principles and their capacity to guide development decision-making processes. The provided results are further examined in a broader discussion section, addressing the relevance of the findings, their limitations, and their practical value for software development teams. Finally, the outcomes of the research project offer insight into the efficiency of Active Persona feedback and suggest directions for future exploration.

## 1.1 Background

The concept of personas for user simulation is recent, introduced first by Alan Cooper in 1999 [2], who proposed the persona concept as a tool for software design. These first personas were static, research-based representations of users, in which Cooper argued that developing detailed archetypal personas could significantly improve product design by providing software designers with a deeper and more nuanced understanding of potential users. Cooper's research emerged in response to a growing challenge within the field of software development during the 1980s and 1990s. In this era, the design process was largely driven by engineers prioritizing technical achievement over user experience. Cooper observed that software engineers often designed for an abstract "user," not a properly defined user that could be referred to as an "elastic user," whose characteristics could be stretched or reshaped to fit any design justification by engineers [9]. As a result, numerous software products reaching the market were technically functional but suffered from poor usability and resulted in user frustration, ultimately failing to meet their needs and expectations. Recognizing this critical disconnect in user-centered design, Cooper developed the concept of personas—detailed, fictional characters based on user research supposed to be representative of existing user types. These meticulously crafted personas encompass the goals of real users, serving as a practical reference for design decisions. By doing so, software designers can group large populations of users with similar needs and interests under a shared identity. This approach enabled product teams to design for broader patterns of behavior and expectations, effectively capturing the needs of potentially millions of people rather than isolated individuals. Subsequent research conducted in 2006 by J. Pruitt and T. Adlin refined the concept of personas by emphasizing the importance of creating rich user profiles that capture behavioral patterns [10]. Their work introduced a structured approach highlighting the role of data-driven personas and the integration of user research to create these rich user profiles that better represent and understand real users. By doing so, their research not only validated Cooper's original idea but also expanded on it by formalizing the process and demonstrating how personas could be systematically generated and applied across design units.

While Cooper's user-centered design approach was considered innovative at the time, some limitations began to surface regarding its traditional implementation. One of the earliest critical analyzes was voiced by Pruitt and Grudin in 2003. While endorsing the concept, they highlighted significant practical challenges in persona implementation within projects [11]. Their research highlighted that if persona design lacked strong implementation methods and support from the organization, these representations of users often didn't impact design choices. Instead of being useful artifacts, they became what they termed "posters on the wall"—a metaphor to capture how personas were frequently created with initial enthusiasm but then abandoned during actual design work, relegated to decorative requirements artifacts while teams reverted to making decisions based on other factors.

Building on these concerns, Chapman and Milham, two Microsoft engineers at the time, published in 2006 a significant critique, *"The Personas' New Clothes,"* which raised fundamental questions about how valid personas truly are [12]. Their acclaimed research clearly explained what they termed the "Representational Paradox" of personas—a paradox that refers to the difficulty in proving whether these fictional user profiles truly reflect real people. The lack of validation in Cooper's and subsequent research work significantly weakened the believability of personas, especially for technical team members who demanded concrete empirical evidence to support design choices.

As the product development environment continued to evolve through the 2000s and 2010s, the criticisms from Pruitt and Grudin, and later Chapman and Milham, gained increasing relevance. The rapid pace of technological innovation further highlighted the fundamental weaknesses of traditional personas. As digital product and service development evolved more rapidly, with more flexibility, Salminen et al. [13] identified the static nature of traditional personas as increasingly problematic. Their research demonstrated that manually crafted personas quickly became outdated in fast-changing environments, failing to capture the latest user behaviors and preferences. This quick obsolescence emphasized the need for more adaptable ways to represent users—approaches that could change and be flexible as user expectations shifted rapidly.

Despite these significant drawbacks, the concept of personas remained a key part of user-centered design. Many organizations understood that although personas were not perfect, they still provided a useful way of making users feel more tangible and anchored empathy and focus within design teams—qualities that data-driven methods often lacked.
The emergence of AI research between 2015 and 2020 revealed breakthroughs in the field of AI-driven personas, both in the private sector—with big tech companies such as Google and Microsoft—and in academic research. These entities began exploring AI-driven personas to simulate user interaction and accurately represent existing users. An example is the research study *Data-Driven Personas* published in 2021, which explored how AI techniques can leverage large-scale data to generate personas that more accurately reflect real user behaviors, with minimal direct user involvement, relying instead on behavioral and interaction data [14]. This research is a strong testament to the potential of integrating AI and data analytics in creating efficient personas that can overcome some of the limitations of traditional, manually crafted user profiles. While such personas played a critical role in humanizing the design process and aligning development with user needs, AI-driven personas offer an interesting approach to the challenges and limitations introduced earlier, allowing for the creation of dynamic, data-driven user representations that are better equipped to adapt to evolving user behaviors, interests, and requirements.

The rise of generative AI models such as ChatGPT, MetaAI, and others has dramatically increased interest in the idea of using AI to better understand user needs and introduce new possibilities. The rapid development of these advanced language models has led to various research studies, such as *Generative AI for Software Practitioners*, which explores how these models are transforming software development by automating repetitive tasks, increasing software practitioners' productivity, and optimizing software testing [15]. The article praises the ability of generative AI to synthesize and contextualize relevant outputs, but the study also discusses the potential risks and limitations of relying on AI-generated outputs. Generative AI can produce plausible sounding but fundamentally unreliable outputs, underscoring the need for careful human validation and expert domain supervision [15]. Therefore, generative AI presents a potentially valuable solution yet still maturing. It might provide a new approach for developing dynamic user personas with the end goal of enhancing software quality assessment and ensuring stronger alignment with evolving user needs and software functionalities.

## 1.2 Scope of the thesis

The scope of this thesis is centered on improving software product development in startup environments by implementing Generative AI-driven Active Personas as a tool for continuous user feedback. The study aims to evaluate the value of feedback generated by Active Personas in assessing software quality and ensuring alignment with evolving user requirements. Rather than exploring the broad spectrum of Generative AI applications in software development, the scope of this research is deliberately constrained to the critical area of user feedback and quality evaluation. The study encompasses the development of Active Personas using generative AI models and their application to two selected open-source applications. These applications (later introduced in Section 3) represent different levels of UI complexity, leading to a comprehensive evaluation of the APs' capabilities across various software environments. While the research evaluates the quality and relevance of AP-generated feedback by comparing it with real user feedback and assessing its alignment with established usability guidelines, it does not extend to the implementation of these changes based on this feedback or deploying APs in production environments. The primary goal is to determine the potential of Active Personas as an alternative to traditional user personas, particularly in startup environments where resources might be limited.

This research addresses a significant gap at the intersection of user-centered design, startup software development practices, and Generative AI-based persona creation. While previous research has examined personas and AI-driven technologies independently, limited scholarly work has explored the practical application of Generative AI technologies in developing active, dynamic personas specifically calibrated to meet the unique feedback requirements of the startup software development process.

## 1.3 Outline

This thesis introduces **five main chapters**. Following the **introduction**, the second chapter presents the **related work**, providing an insightful review of the existing literature regarding user participation in software product development, the concept of Active Personas, and Generative AI technologies for user modeling. This chapter identifies the gaps in current research and sets the foundation for this research project.
The third chapter describes the **selected research methodology**, outlining the specific research questions that guide this study, as well as detailing the design and implementation of Active Personas using generative AI models such as Meta AI and Open AI. This chapter also outlines the selection criteria for the two open-source applications used in the evaluation, the framework implemented to simulate user interactions, and the process for assessing the quality and relevance of the Active Personas' feedback generated.
Chapter 4 presents the **results and analysis**, introducing the feedback produced by Active Personas during their simulated interactions, analyzing its contextual alignment with user-centered design principles, evaluating their performance, and comparing the generated feedback with existing real user feedback to assess its validity.
The fifth chapter provides a **discussion** by analyzing the findings, addressing the potential of Active Personas as a tool for improving software development. It critically examines the strengths, challenges, and limitations of the approach.
Chapter 6 **concludes** the research project by summarizing key contributions, reflecting on the original claims and research questions, and exploring paths for future investigation into the application of Active Personas in software development.

# 2 RELATED WORK

## 2.1 On the content

The following sections aim to summarize and contextualize related research work that aligns with or contributes to the research of the Active Persona concept, as well as the usage of Generative AI in new product development and user experience design. The review of these related research papers is directly linked to the subject of this thesis and provides information on the theoretical foundations and practical applications necessary to support and guide the study.

### 2.1.1 The important role of user development in software development success

User involvement is another topic related to this research project. As the key issue identified, several research papers investigated the important role of user involvement in the success of a software product. One study that summarizes the complexities and nuances of user involvement in software development success is *User Involvement in Software Development: The Good, the Bad, and the Ugly* [3]. The main objective of this paper is to investigate the role of user involvement, both positive and negative throughout the software development process. This study is empirical qualitative research that categorizes the challenges of user involvement into three categories: "the good" beneficial, "the bad" problematic, and "the ugly". From the collection of qualitative data obtained through interviews with stakeholders in the field of software such as users but also software developers and project managers. Based on the analysis of these data, the authors identified recurrent themes that express the multifaceted nature of user involvement.

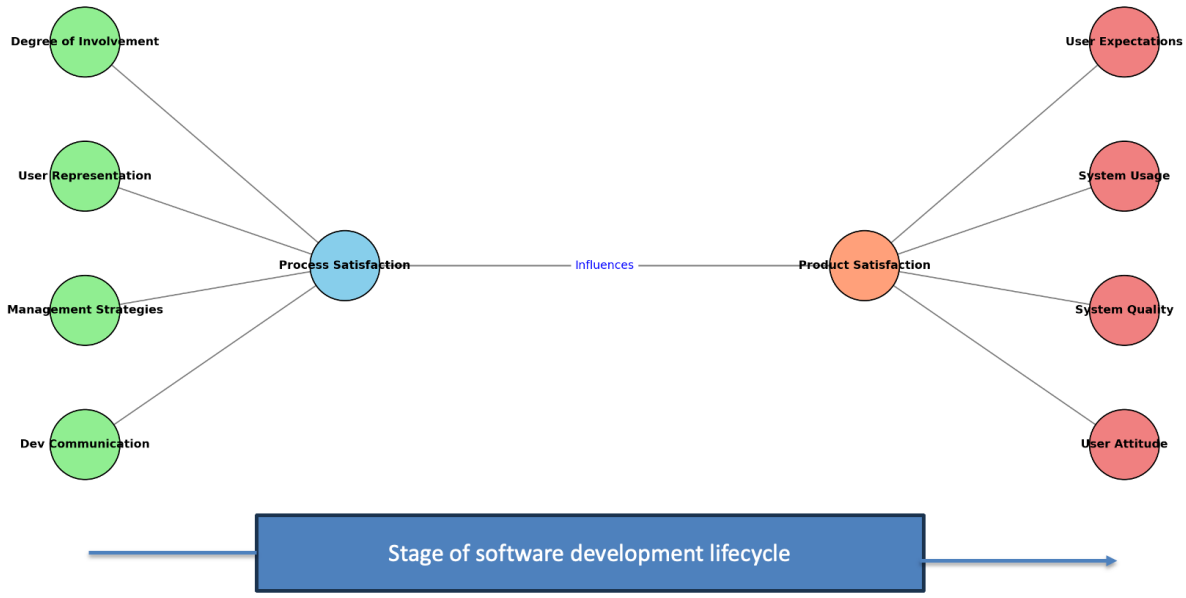



Figure 1: Users' satisfaction/dissatisfaction with their involvement in software development redraw from [3]

In Figure 1, the authors defined user satisfaction as tightly linked to their satisfaction with the involvement process, defined as a cyclical relationship between the process and product that influence itself. They identified through that framework defining key factors impacting the satisfaction of end users throughout the stages of the software development lifecycle:
Degree and levels of involvement, user representation, management strategies, user-developer communication. These factors contribute greatly to the evaluation of the user involvement process that influences the end-user with the final product in terms of expectations, usage, quality and behaviors

toward the system (product). This framework challenges the established claims that user involvement itself necessarily leads to better project quality and success. By introducing those factors (blue on Figure 1) the authors demonstrate the complex relationship between user own satisfaction during the involvement process with their eventual satisfaction with the final system factors (orange on Figure 1). The cyclical nature of the framework reveals how negative experience during the involvement phase can hinder the system adoption, categorized as “the bad”. The opposite is also true, positive engagement can favor product quality and adoption by users and be categorized as “the good”. If some issues and challenges are not addressed early in the process such as misaligned expectations or poor management strategies, it can escalate to “the ugly” [3]

This research is important and relates to our own exploration of dynamic and generative persona. It highlights the importance of the user involvement process and its relationship to the final user satisfaction. The study supports the thesis by affirming that user involvement alone is not sufficient and other factors are required such as meaningful engagement, expertise and representation. Obstacles that APs will be approach differently by implementing generative AI.
Another research uses the framework created in Figure 1 as a solid foundation for exploring the broad role of user engagement during the software development lifecycle [16]. The research endorses the claims established about the influence on user participation by technological and social factors, which consequently affects software quality and development success. The research paper *User Involvement in Software Development: Factors, Challenges, and Recommendations* not only validates the framework on user satisfaction and dissatisfaction influences [3] but also contributes to the ongoing investigation into how user participation affects software development outcomes by advocating adaptive and flexible development methodologies [16].

### 2.1.2 AI as a creation tool for Co-Design framework

The research article *Exploring the Integration of Artificial Intelligence in Co-Design: Framework for Designer* analyzed the potential for artificial intelligence LLMs to assist designers during the co-design phase [17]. Instead of positioning AI as a replacement of actual designer, the research introduces a framework based on peer reviewed studies that identified key opportunities where AI can be applied and have a significant impact on the design process. The identified key opportunities can be joined under 4 areas: idea generation, user feedback interpretation, and iterative prototyping. In all those areas AI was demonstrating promising abilities. The authors emphasize the capabilities of AI in improving both technical processes and triggering emotional engagement notably in applications such as Co-Design.

As this article does not mention Active Persona directly, it still provides valuable knowledge about how generative AI can improve early-stage design by supporting decision-making, in this case improving overall functionality and reliability. The findings suggest various directions for building Generative-AI agents that support creativity, flexibility and user-centered design process.

### 2.1.3 A Vision on Active Persona

The concept of Active persona is relatively recent concept in the field of user-centered design and software product development. The research article presented in [18] introduces an innovative model that addresses persistent challenges related to user participation and feedback collection throughout new product development (NPD), particularly in resource-constrained and fast-paced environments such as software startups. This work can be considered a key starting point for understanding the concept of Active Personas.

The research paper starts by exploring the shortcomings of traditional personas, although they are widely used in requirements engineering, usability testing, their static nature is often pointed out as a major shortcoming that often-limited user representation and are subject to misinterpretation and limitations of user representativeness. To overcomes these limitations, the authors introduce the concept of Active Persona, AP are AI powered interactive entitles that integrate user personas and requirements engineering artifacts to generate contextual feedback through multi-modal interactions. As opposed to static personas that are defined as reference documents, APs implements generative AI to simulate user interaction and provide dynamic feedback during the software development phase [18]. APs integrate various requirements engineering artifacts to build a cohesive AI power agent, including user personas representing actual user characteristics and needs, goals, motivation. But also, user scenarios as contextual narratives describing how user interact with the systems, user stories that will play the role of functional requirements written from user perspective. These artifacts are processed through generative AI models with modern techniques such as Retrieval-Augmented Generation (RAG) that combines provided data with generative capabilities. Among those capabilities, multi-modal processing that handles text, voice, image and video inputs and outputs.

APs behave as an AI tool that helps check if requirements are met and that enable developers to process iterative internal experimentation by testing prototypes and gather immediate feedback without direct external user dependencies. APs can generate contextual response by providing relevant feedback base on the provided user characteristics and requirements artifacts, simulate real user interaction through multiple input types and reduce the extra effort, time, and resources required to verify that the product meets the needs and expectations of real users.

The rapid innovation around Generative AI technologies those past three years, provided a solid technical foundation required to concretize the APs concept. Generative AI, especially through the prism of large language models and multimodal systems enables the generation of convincing text and outputs that are accurate and qualitative. Consequently, those innovations empower APs to generate simulated user interactions during the initial product development phase where fast interaction and hypothesis testing are essential.

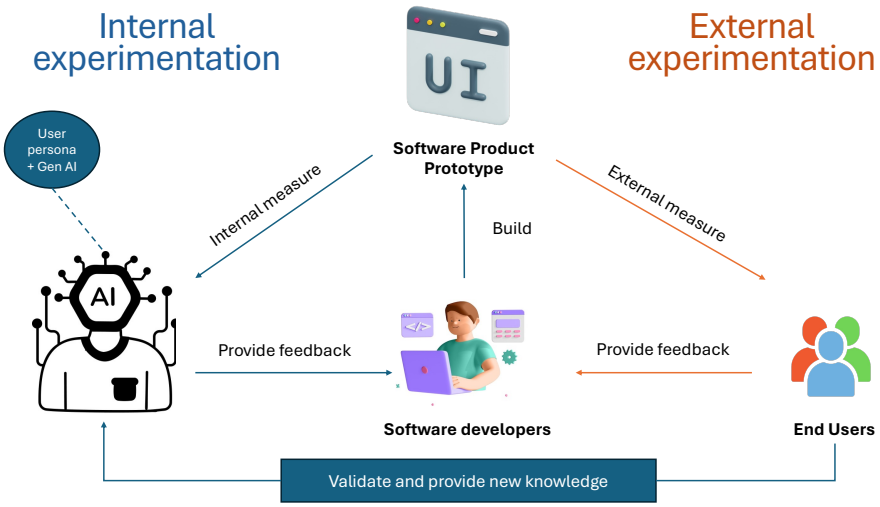



Figure 2 Active Personas Concept

In Figure 2 extracted from the research article [18], illustrate the Active Persona process and workflow within the context of software product development. On the left side of Figure 2, the internal view of a project structure is depicted and introduces how APs are created. They are created by combining already established user persona and GenAI including (RAG) techniques. Those newly crafted APs are then used in internal experimentation by interacting with the prototype and creating internal measures for providing insight on the interaction. The APs will then provide multi-modal feedback (through text, voice or visual content) to the development teams. The development team will use this feedback loop to learn from the APs and refine the prototype iteratively before delivering the product to market. On the right side of Figure 2, the prototype is introduced to real users in external experimentation phases, developers collect feedback from the real word. The feedback collected from real world users is not only used to guide the development of the software but also serve as reference point for the APs to align with evolving user needs, understanding of the product and creating a continuous cycle of improvement. The main purpose of the complete feedback loop (internal and external) is to bridge the gaps in user involvement during the development process.

The article surely introduces the concept of Active Person and establishes their potential role in new product development; it does not provide technical details on how APs should implement or how they are expected to behave in specific scenarios. The proposed approach is purely conceptual, establishing Aps as a vision to address challenges in user participation by leveraging generative AI and implementing requirement engineering artifacts. The article is falling short of specifying the underlying architecture of APs, which interaction models might be the optimal approach to evaluate a prototype and generate valuable feedback. The article rather serves as a theoretical blueprint for a potential implementation of a prototype, highlighting the potential applications and expected benefits for practitioners building software, without offering tangible proof of concept. The authors acknowledge those limitations by stating in their article that their next step involves building an initial prototype and conducting case studies to explore the concept viability in practice. This research proves to be a valuable foundation and a starting point for our research, which aim to further develop and implement an AP prototype and evaluate it in practical settings.

### 2.1.4 Generative AI Applications in User Experience Evaluation

Another recent study investigates the implementation of LLMs and personas to facilitate adaptive user experience (UX) design [19]. The author starts by also acknowledging the limitations of traditional persona in UX methodologies, especially pointing out their static nature and inabilities to have impact on interfaces during the development process. The research proposition is based on the use of a framework that centers LLMs for improved UX development by first, understanding adaptive UX components then analyzing the role of personas in Adaptive UI/UX, evaluating LLM prompt techniques, and finally framework development and evaluation. This approach reinforces the growing enthusiasm for AI-driven UI/UX design and further mark the shift toward more intelligent and flexible user-centered interface development processes (Figure 3).
This research work is particularly relevant to our subject, especially the reflections around using LLMs as tools for personalizing UX and guide adaptive UI. The author claims that LLMs when paired with a well-structured and rich persona, can dynamically adapt UI components such as color scheme and layout based on user preferences. The authors identified **self** (user own preferences), **world** (environment, contextual information), **optimal** (best possible interface), and **utility** (value of design choice) as different types of interference that LLMs can analyze to support adaptive UX [19]. This research proposal resonated with our goal of developing Active Persona, persona that are dynamic and implement Generative AI and engage actively in UI/UX evaluation and contextual feedback generation.

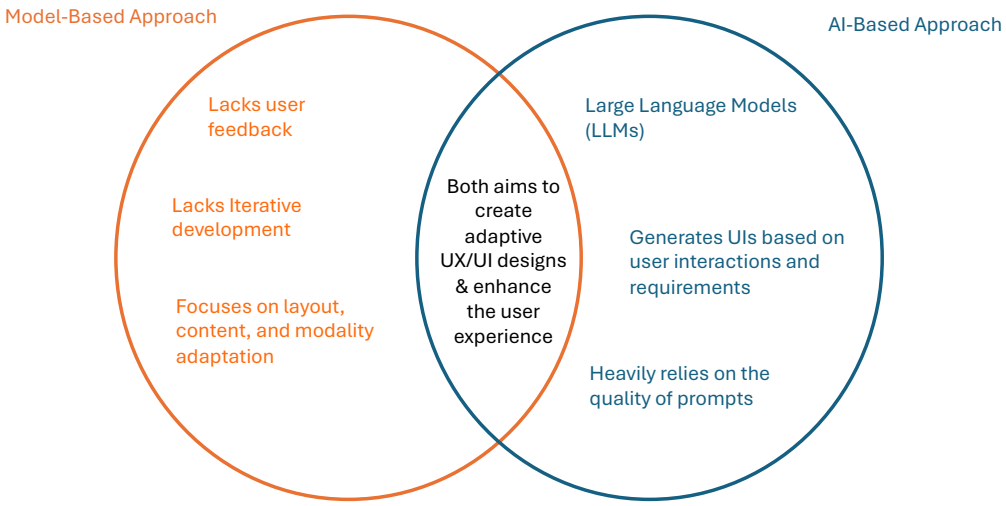



*Figure 3 : Adaptive UI/UX Approches : Model vs. AI*

While both research project aim at enhancing user-centric design process through AI, key differences can be observed. Large language models (LLMs) have been applied to adaptive UI and UX component generation using persona-guided design strategies [19]. As opposed, our work focused on using transforming static persona into powered generative AI persona as a tool to simulate interactive user and generate feedback that evaluates a software product. Whereas their framework proposed modifying UI and UX components (such as layout, typography) based on user preference with a focus on evolving actual UI/UX components by generating new adapted ones.

To synthesize, this research work shares many commonalities with our own research project and provides a compelling case for the usage of LLMs in adapted UI/UX generation. It serves as evidence that our research subject is relevant and being conducted at a particularly convenient time. The distinction between the two-research project highlights the novelty of our contribution, positioning APs not as design generators but as efficient evaluators that can generate feedback based on actual user needs.

Another contribution investigates the potential of large language models (LLMs) in supporting user interface development. The study *Unlocking Adaptive User Experience with Generative AI* explores how ChatGPT-3 can be used to automate persona generation using limited data and how those personas can, in turn, generate and refine user interface design ideas [20]. While their research includes a limited amount of data with 37 survey answers and 4 semi-structured interviews, they've managed to produce valuable artifacts that combined both AI generated and traditional manual outputs [20].
The personas and interface concept generated by Artificial intelligence were used in a comparative analysis beside human manually crafted artifacts, they were compared in terms of quality, usefulness in guiding UX decisions.
The findings of the comparative analysis reveal that ChatGPT was able to produce qualitative personas comparable to the one crafted by humans in terms of content and structure. The authors point out the efficiency of the LLM in generate accurate personas, accurately incorporating existing behavioral pattern with limited amount of data while achieving important gains in time efficiency. The crafted interface generated by AI presented in the research paper is solid evidence that the GPT-3 could adapt to content to different personas, which is going to be a key aspect in our own research project.

On the limitation, the selected LLM for the experimentation (GPT-3) was prone to unsupported assumption according to the authors that also emphasis the necessity of having human supervision regarding the validity of the LLM outputs. The quality of the output provided by the AI was identified as strongly dependent to prompt quality. Unfocused or poorly designed prompt often led to misalignment and irrelevant outputs. The research conclusion emphasis that the heavy reliance on prompt engineering in the study introduces a threat its validity [20].
Despite those limitations, the authors highlight the increased time efficiency in the design process that generative AI solution offers and the benefits of implementing such a solution as a collaborative tool rather than a standalone solution aimed at replacing traditional UX designers.

This research is relevant to the present thesis and matches the concept of Active Persona, esspecially with the issues they are trying to solve such as the difficulty of collecting representative user feedback. Their findings clearly establish a need for further investigations regarding the usage of Generative AI models (other than OpenAI LLM) in user experience evaluation. The identified limitations regarding prompt engineering and AI assumptions might be limitations that will be encountered during the experiment of this research project. The usage of minimal data required for LLM output is also insightful and relates to the concept of APs generating feedback without requiring constant human input.

# 3 METHOD

## 3.1 Aim and objectives

This research aims to validate the concept of Active Personas (APs) as an effective tool for generating contextual software development feedback. By systematically analyzing the efficiency of AI generated Active Personas, the study will thoroughly examine their ability to produce meaningful insights that match or exceed real user feedback.

**Objectives:**

- Development and implementation of Active personas using Generative AI models including Meta AI Llama 3 and Open AI.
- Apply the developed APs to selected open-source applications with varying UI complexity to simulate user interaction and generate relevant feedback observation after interacting with the UI part of it.
- Analyzing the quality and contextual significance of AI generated feedback and comparing it to real-world user behaviors by using industry accessibility guidelines.
- Analyze the limitations and shortcomings area of APs and identify areas for improvement.

The defined objectives address a critical gap in software development environments where resource limitations often constraints comprehensive user testing. By focusing on the development and validation of Active personas as alternatives to traditional user feedback techniques, the study not only demonstrates the achievability of using AI-driven user simulation to provide meaningful feedback but also demonstrate the potential of such an approach for development teams seeking innovative methods to gather meaningful user experience data without extensive resource allocation.
In Addition, those objectives seek to highlight specific areas where APs may excel in identifying usability issues but also situations where their insights may fall short compared to real user testing.

## 3.2 Research questions

Defining relevant and answerable research questions is a fundamental landmark in establishing the structure and scope of this research project. In this study, which investigates the use of APs as a tool for generating user feedback in software development, the following research questions are designed to explore both the value and limitations of this innovative approach. The aims (defined in section 4.1) go beyond simple validation and encourage discussion and reflection on how APs can influence software quality and enhance user alignment, especially in the field of tech startups where traditional user engagement methods may be time-consuming and costly [21].

A good research question should be open enough and allow a deep analysis and interpretation but also be defined in a way that required focus to be meaningfully answered through systematic investigation. With that in mind, the primary research question (**RQ1**) was defined to assess the degree to which APs feedback can come close to real user needs and contribute to improving product quality.

**RQ1 (Primary): To what degree can AI-powered Active Personas generate contextual user feedback that aligns with real user needs and improves software product quality?**

This question does not seek a simple "yes / no" type of answer but rather introduce a qualitative and comparative exploration across different use cases and frameworks. RQ1 seeks to understand not only if APs can generate feedback but also to examine if the generated feedback is contextually relevant, practicable and aligned with establish usability principles.
The secondary research question (**RQ2**) is complementary to the first, by addressing the potential challenges and constraints identified during the development and application of APs.

**RQ2 (Secondary): What are the systematic limitations and potential areas for improvement in AI-driven user interaction simulation?**

This question furthers deeper analysis and critical assessment of APs by exploring their actual shortcomings. It encourages reflection on how these issues might be addressed in future iterations, potentially through more advanced AI models or a hybrid approach combining real and generated feedback. Combined, these two research questions (RQ1 & RQ2) are designed to provide a balanced examination of Active Personas, exploring their potential value for generating aligned user feedback but also their intrinsic limitations. These two approaches create a framework for critical evaluation that goes beyond simple validation, offering practical insights for tech startups considering using Gen AI as an alternative tool to traditional user testing methods.

## 3.3 Research methodology

The research project will follow the Design Science Research (DSR) methodology to create Active Personas with the objective of generating contextual feedback on selected codebases. By choosing the DSR framework as the research methodology for this project, the aim is to create practical artifacts that address real-world challenges in the product development cycle. The DSR framework will follow its defined steps, known as problem identification, artifact conceptualization, design and development, demonstration, evaluation, and communication of research [22].

The study developed 3 AI-powered Active personas as its primary artifacts. Which emulate user interaction by generating feedback on selected mobile applications. These personas will be representative of real daily users using these mobile applications. The goal of these artifacts is to serve as tools to investigate if Gen-AI can be exploited as a reliable tool within user-centric design process.

### 3.3.1 Artifact Design: Active Persona

After research conducted on the type of daily user that mostly use the selected mobile apps (presented in section 4.3.2), two personas were designed to represents the diversity of user profiles and digital habits.

**Table 1- Persona A: Alex – The Fast-Adapting User**

| **Name** | Alex |
|---|---|
| **Age** | 24 years old |
| **Background** | Undergraduate student in marketing |
| **Technology knowledge** | High, very comfortable with mobile and computer apps, use them extensively for work and private life for various purpose, embrace new technology rapidly. |
| **Requirement in a product** | Robust (can handle heavy workload), Efficient, minimal distraction, feature focused, privacy encryption is a plus. |

| | |
|---|---|
| **Elements in a product to avoid** | Disorganized interface, long registration settings, bugs and crashes. |
| **Device used** | Android user (high end phone), latest OS version |
| **Used app** | What's App, Telegram |

Alex represents Gen Z users, comfortable using new technology. He represents users that are familiar with other popular live communication apps such as WhatsApp and Telegram. He is comfortable in digital ecosystems and likes software that values privacy. His profile makes him suitable for testing how well the UI introduces advanced features and balances them with usability for new users experienced with similar messaging apps. His expectations are high regarding user experience—he wants a mobile app that can keep up with him, be pleasant to look at, easy to navigate, and modern. He is likely to criticize UI shortcomings and inconsistent design.

**Table 2- Persona B: Geraldine – The Casual user**

| | |
|---|---|
| **Name** | Geraldine |
| **Age** | 61 years old |
| **Background** | Administrative Clerk in public health care service, no background with I.T or computer interaction. |
| **Technology knowledge** | Moderate, comfortable using popular apps like Gmail, What's App, make video call to her family but avoid anything "too engineered" or overly complex. |
| **Requirement in a product** | Simplicity, clarity, Ease of use, secure feeling when interacting with the app. Consistent Navigation. Minimal step to complete a Tasks. |
| **Elements in a product to avoid** | Confusing user flow, technical terms, Hidden UI elements or too many steps for completing basic tasks. |
| **Device used** | Android user (Mid-range phone), often use accessibility features option like making the font a bit bigger. |
| **Used app** | Firefox, Bitwarden |

Persona B is an interesting user that is often overlooked in the design process of a software product, since this type of user are often considered function-oriented and cautious. Geraldine embodies the demographic of aging but active user who engage with technology daily, out of necessity rather than leisure or entertainment. Her interactions with Bitwarden provide insights on the app clarity, accessibility and cognitive load required for basic task such as secure logins set up and vault management. This persona will play an important role in evaluating Bitwarden functional clarity thereby ensuring that trust and ease of use are never compromised.
"Geraldines" are not early adoption users but represent a sizable and growing segment of the user base that are often overlooked [23]. Ignoring their needs could be consequential to the overall application quality and a missing opportunity to build efficient usability and trust into the product.

A third persona will be used to provide technical feedback with a codebase approach, while the two first persona (A & B) focus on simulating user interactions with the UI. The third persona (C) is not representing a specific type of user but instead to act as an expert in code quality analysis powered by AI models. The purpose of Persona C is to examine the code quality attribute of the codebase to provide a deep analysis on the quality of the application, provide areas of improvement, strengths but also weaknesses. Using this persona will be key in providing feedback on more localized technical issues that may not be visible through user interaction alone.

**Table 3- Persona C: The Expert Code Quality Analyst**

| | |
|---|---|
| **Role** | Expert code quality analyst |
| **Type** | Functional, non-behavioral expert. |
| **Purpose** | Evaluate the quality of selected Android apps, investigate specific issues. |
| **Focus Area** | Maintainability, Usability, Readability, Robustness. Accessibility. |
| **Expertise** | Software architecture, static analysis, adherence to coding rules and principles. |
| **Output** | Strength, weaknesses and localized technical feedback. |

Representing different age groups and user behaviors, these 3 personas will be used as key components in the Design Science Research framework. When paired with the selected generative AI models (presented in section 4.3.3), realistic feedback and rich interaction with the selected mobile applications are expected. The combination will enhance the validity of the experiments by reflecting on diverse usability needs and user expectations.

### 3.3.2 Application and codebase selection

Two open-source Android applications were chosen with the end goal of assessing how well APs produce contextual feedback: **Bitwarden** and **Signal**.

- **Bitwarden (2024.10.0)**:
  Bitwarden is an android password management application popular among users looking for an alternative to save their important password in a reliable encrypted way. The app includes multiple features such as credential input, password generator and encrypted vault, the app also implements a secure web browser feature. The application started as an open-source project in 2016, the codebase is available for everyone on their official GitHub page [24], today the application counts approximately 5 million downloads. The user interface can be considered straightforward and well garnished, focused on making the security features accessible for most users. It serves as an ideal testing case for APs on evaluating the clarity, accessibility and assistance of users in navigation and understanding of specific features. The codebase located on GitHub is composed of approximately 99.8% of Kotlin file and uses a modular architecture that follows the Model-View-View-Model, applying a separation of concern that facilitates the distinction between UI components and the logic. This architecture should facilitate the evaluation of the codebase by Persona C.

- **Signal (7.38.5)**:
  Signal is on the other hand, is a much more popular app with approximately 100 million downloads. Signal is an encrypted messaging application that value privacy with a minimalistic user interface. As opposed to Bitwarden, the app quality encompasses speed, clarity and try to keep cognitive overload as low as possible. Making it a good contender to evaluate how well APs respond in a minimalistic environment and commutation efficiency are put first. Signal is also an open-source application, with the code source available on their official GitHub page [25]. The software architecture of the application implements a variation of the Model View Presenter (MVP) pattern, an architecture that are famously found on messaging services since it facilitates the real time messaging capabilities of the app while making sure security appear unbreachable. This application was selected for its straightforward UI and low barrier user

scenario with everyday usage, used by millions of users daily which should point out interesting user feedback regarding the UI but also the reliability and accessibility of the app.

### 3.3.3 Generative AI model selection

In accordance with the DSR framework methodology, this study implements Generative AI models as the core technology behind Active Persona. Multiples Generative AI were selected based on their suitability regarding the selected mobile app user interface but also regarding the codebase analysis. The models will be run in different environment with different purpose: the first implementation (1) of APs will be running via a python-based API library for the Open AI model (through the cloud), the second configuration (2) will require Ollama. Ollama is an open-source framework used to run AI models locally, each model is downloaded and is being executed locally, no data are shared through the cloud.

**Personas A & B – UI Evaluation Based on Visual Input**

Persona A and B was designed with the goal of simulating user interaction through the UI of Bitwarden and Signal. The evaluation will be operated by loading those personas as APs and providing screenshots from the apps running to analyze them and generate feedback.
These crafted personas will be coupled with a Gen AI model, in that case **OpenAI's GPT-4o** that will be implemented using a Python external library. The feedback generated will be given instruction through a prompt to analyze the screenshots provided by analyzing usability quality such as clarity, button layout, colors chart, ease of use. The prompt might be adjusted to investigate specific issues, for instance a specific clicking button that is not working.
GPT-4o was selected for this initial setting (1) mainly due to its ability to be a model that can process images and text, as opposed to other Large Language Model (LLM) that are limited to process and generate text only (2). Although it is not an open-source solution and require payment for access.

**Persona C – Codebase Evaluation with Text-Based Models**

The second setting (2) focuses more on a technical review providing a deep and more tangible analysis of the codebase of Bitwarden and Signal. Focusing on code quality attributes such as readability, accessibility and usability.
To perform the analysis and generate the feedback from it, the following open-source LLM were chosen:

- **Meta AI Llama3 (8B)**: Introduced by Meta in 2024 as a language model with more than 8 billion parameters.
- **Mistral (latest version)**: Launch to market in 2023 by the French company Mistral AI, renowned for its performance and great efficiency.
- **DeepSeek (R1)**: Published on the AppStore in 2025 by a Chinese company of the same name, DeepSeek is gaining momentum with growing user adoption and is viewed as an emerging competitor to U.S LLMs. Also praised for its efficiency.

These models are only text based and will run locally through Ollama, which is a factor to consider as startups company don't want to rely on external API dependency and often want to prioritize offline access, data privacy and cost saving solution.

To measure their practical performance, each model will be measure by their execution time while they are performing their evaluation. This measurement will complete the evaluation in determining not only the quality but also the efficiency of the APs feedback generation concept and therefore being relatable to real-world application, especially for environment were resources and time are important factors.

### 3.3.4 DSR implementation stages

As mentioned before, the research project adopts the DSR framework methodology to design APs that can generate contextual feedback on the selected mobile applications. The main objective behind the choice of this research methodology for this project is to design practical artifacts that addresses tangible challenges in the product development cycle. Challenges that are recognized and shared by practitioners and researchers.
The DSR framework incorporate 6 defined phases, in the following order: **Identification of the problem**, **design and development of artifacts**, **demonstration by using the artifact to solve the problem**, **evaluation of the solution**, **reflection & communication of the research** [22].

**Problem Identification**:

A recurring challenge noticed in the early development phase of a software product is the difficulty of involving the end-user to ensure that the final product meets their needs. Especially for small organizations of type startup, where economic resources and time are constrained. The lack of user involvement and direct user feedback can lead to usability issues and user barriers that can ultimately lead to product failure [26]. Therefore, there is a clear identified need for a scalable and innovative approach that can accurately represents actual user feedback early in the development lifecycle.
This research aims to address this issue by exploring the potential of Generative AI through the concept of Active Persona to generate contextual feedback on specific software mobile applications.

**Artifact Conceptualization**:

The main artifact designed is a system of Active Personas that are powered by generative AI models with the objective of generating meaningful contextual feedback. This system has been designed to represent diverse user perspectives and technical knowledge, delivering insightful evaluation on both the user experience (UI) and the source code quality. The APs implement three distinct personas, each carefully crafted to match actual user that are representative (introduced in section 4.3.3).

- **Persona A:** A younger, task and UI focus.
- **Persona B**: An older, accessibility-conscious user, prioritizing clarity and usability.
- **Persona C**: An expert analyzing codebase quality.

**Design and Development**

The artifact will be composed of the developed personas (A, B, C) coupled with a generative AI model which will make them Active persona capable of interacting with the selected mobile application, Bitwarden and Signal.
Persona A and B implements Open AI's GPT-4o with a computer vision approach, the model will be implemented using Python official Open AI library.
Persona C will implement text base generative AI model to analyze application code using Meta AI's Llama3, Mistral (latest version) and DeepSeek (R1). The models will be downloaded and executed locally. They will be implemented into Persona C using Python as programming language.

**Demonstration**:

Each respective Active Persona will interact with the selected application,

- Persona A → Signal: Receive application screenshots and simulate user interactions based on provided tasks.
- Persona B → Bitwarden: Receive application screenshots and simulate user interactions based on provided tasks.
- Persona C → Bitwarden: Reviews selected parts of the source code repositories.

After interacting with the app, all APs will generate feedback based on their specific focus. The feedback will then be collected and categorized for further comparison and analysis.

**Evaluation**:

The quality and contextual accuracy of the generated feedback will be evaluated manually but in alignment with official ISO guidelines. The qualitative evaluation will be an assessment step to ensure that the generated feedback is contextually accurate, coherent and usable with particular attention to identifying and eliminating potential AI generated hallucinations. The evaluation will then be guided by established principles from the International Standard ISO 9241-110:2020 and ISO 9241- 112:2017. These guidelines will play a key role in assessing whether AI-generated feedback aligns with well-established usability principles. Table 4 presents the selected principles from both guidelines, the following principles were selected for their relevance in assessing the clarity, consistency and user first approach, qualities that users like to find in software product.

| Principle | Guideline | Reference |
|---|---|---|
| **Consistency** | **Guidance related to using presentation attributes:**<br><br>1) **6.3.3.1** Items of information that are logically dissimilar should be presented differently (e.g. using different presentation attriibutes) to make their differences obvious.<br><br>NOTE: This can involve locations, visual shapes, sizes, colours, typefaces, etc.. ; auditory order of presentation, volumes, tones, frequencies, etc ..; tactile locations, shapes, vibrations, pressures, etc.<br><br>**Guidance related to structuring of presented information**<br><br>2) **6.6.2.3** Where attributes are used to encode properties of pieces of information, they should be used consistently throughout the interactive system, or at least throughout a single state of the interactive system.<br><br>NOTE : Where encoding attributes are used differently in different states of an interactive system, it is important that changes to the state of the interactive system that also change the encoding are made clear to the user. | ISO 9241-112:2017(E) |

| | | |
|---|---|---|
| **Controllability** | The interactive system allows the user to maintain control of the user interface and the interactions, including the speed and sequence and individualization of the user-system interaction. | ISO 9241-110:2020 |
| **User engagement** | The interactive system presents functions and information in an inviting and motivating manner supporting continued interaction with the system. | ISO 9241-110:2020 |
| **Freedom Of Distraction** | **Guidance on minimizing and avoiding distractions**<br><br>**6.2.1 Principle of freedom from distraction**<br><br>Presented information is free from distractions if the information is presented so that required information [1] will be perceived without other presented information interfering with its perception. Distractions from a user's point of view can result [2] from both distracting events and information overload.<br><br>Freedom from distraction involves guidance related to<br><br>a) avoiding distractions, and b) minimizing distractions. | ISO 9241-112:2017(E) |
| **Conciseness** | **Guidance related to conciseness of content**<br><br>**6.5.2.1** Minimalism and simplicity should be used to present information concisely. | ISO9241-112:2017(E) |

*Table 4: Selected Guidelines Principles*

The execution time of each generated model will be measured and compare to evaluate their efficiency at generating feedback. This measurement will be important in ensuring the viability of each model in real-case scenario and their suitability in specific environment.
Another evaluation of the AP feedback will be the comparison with real user feedback. User feedback from the Google Play store will be retrieved for the specific versions of the application under review. The comparison will be key in highlighting the similarities or gaps between real user concerns and APs feedback.

**Communication**:

The findings of the evaluation will be presented in the results and discussion chapters (5 & 6) of this report through a comparative analysis detailing the strengths, limitations and practical implication of relying on APs generated feedback in the development process. The communication of the results will emphasize on the context of startup environment with a focus on how APs might offer an innovative, resource efficient solution to support iterative product improvement.

**Flow diagram of the DSR Method:**

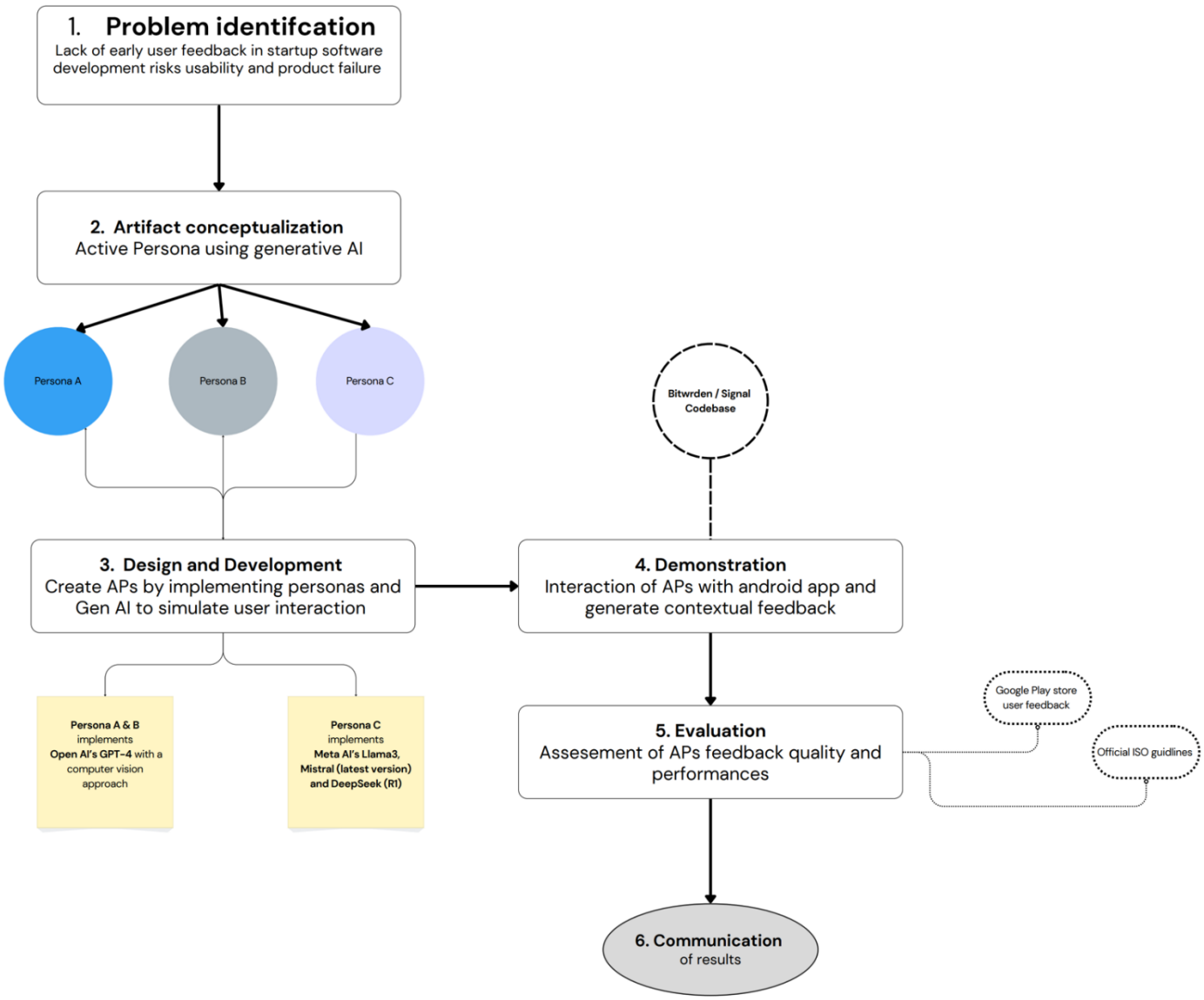



*Figure 4: Flow Diagram of the DSR framework*

By adopting the DSR as a research methodology, a clear and structured path to design and implementation has been established to evaluate the concept of APs as artifacts. This approach is beneficial on many aspects such as, clearly identifying the problem to solve and which resources are going to be employed and design to address it effectively.

#### 3.3.4.1 Artifact Objectives and Requirements

As required by the Design Science Research Methodology (DSRM), this section introduces the objectives and design requirements of the Active Persona artifacts. These objectives serve as references for both the design choices established in the artifact development phase and the criteria defined to evaluate its success. The purpose of defining the objectives and requirements is to ensures the artifacts fulfills its intended purpose and directly address the problem identified in Section 1.

##### 3.3.4.1.1 Objectives of the Active Persona Artifact

The main objective of the DSRM artifacts is to generate contextual user feedback. The objective will be reach through the implementation of generative AI agents, designed as Active Persona. Those artifacts will interact will software product, the core objectives of the DSRM can be found in the following table (Table 5):

| Number | Objective | Description |
|---|---|---|
| **1** | **Contextual Feedback Generation** | The AP should generate feedback that is aligned with established with renowned usability defined in Section 3.3.4. Adhering to the selected usability principles will ensure feedback validity and will provide a theoretically grounded and consistent evaluation. |
| **2** | **Persona Implementation** | Each artifact (AP) must be able to generate feedback that reflect the implemented user persona demography and behavioral profile (e.g., a tech-savvy user vs. an older, casual user). The final output must be influence by the selected personas. |
| **3** | **Prompt Processing and Accuracy** | The APs must be able to process structured prompt provided that include text and in some cases screenshots, and they should be able to provide generated feedback within a reasonable timeframe. |
| **4** | **Alignment with Real-user Feedback** | The generated feedback should touch issues, correspond or reasonably match real-world feedback from actual user (feedback collected from the Goole Play Store), providing a validation from authentic usability concerns. |
| **5** | **Systematic Feedback Analysis & Actionability** | The generated feedback must be systematically analyzed and categorized using the corresponding usability standards. The goal is to assess the contextual accuracy of the feedback but also process a diagnostic on quality and potential on directly inform on design improvements. |
| **6** | **Minimal Resource Dependency** | The artifact should run locally by using open-source Ai models (e.g., Mistral, LLaMA3) to minimize dependency on commercial APIs. This objective will support the use of AP in resource-constrained environment such as startups. |

Table 5 Artifact Objectives

3.3.4.1.2 Requirements Derived from Objectives

| Requirement | Description | Type |
|---|---|---|
| **Feedback alignment with ISO guidelines** | The generated feedback must adhere to established international usability principles (e.g., ISO 9241-112:2017(E) / ISO 9241-110:2020) | Functional |
| **Feedback matching real user concerns** | Generated feedback should correspond to or reasonably match actual user concerns observed in real-world data collected from the Google Play Store for the specific version of the selected app presented in Section 3.3.2. | Functional |
| **Feedback must reflect Persona behavioral profile** | Must produce feedback that accurately reflects the demographic and behavioral characteristics of the simulated user persona defined in Section 3.3.1. | Functional |
| **Implement local model approach (where possible)** | The created artifacts must include a solution that can be run locally and use open-source models that will enable private deployments. | Non-Functional |
| **APs must process multimodal inputs** | Enables multimodal evaluation of app UI/code with screenshots provided as. Jpeg or Png files. Prompt must be structured with persona implemented as an external text file. | Functional |
| **Feedback generation within reasonable timeframe** | For iterative product development that only include image-based UI task, the feedback must be generated in less than 5 minutes (<5min) and for code base analysis the time should not exceed 1 hour (<1 hour). Running time above those specific timeframe will be considered impractical/ ineffective, since it will impact the agility and flexibility required for continuous, early-stage product development cycles by startups. | Functional |

Table 6 Artifacts Requirements

3.3.4.1.3 Evaluation Alignment

The defined objectives in Section 3.3.4.1.1 will be explicitly used in the evaluation phase of the artifacts (Section 4.3). The quality of the Active Persona generated feedback will be assessed through: ISO principles mapping, comparison to real user reviews, consistency across different models, and their running time to generate feedback. By establishing these objectives and requirements for the artifacts, the AP implementation is strategically aligned with the research goals and positioned for robust evaluation.

## 3.4 Validity and Reliability

To assess the validity of the DSR demonstration, this study will also incorporate real user reviews retrieved from the Google Play Store, specifically for the same app versions used during testing. A script was developed in Python to scrap all the reviews by using the python library *google-play-scraper*. Snippets of the reviews can be found in *Appendix 8.6* for Bitwarden android app and *Appendix 8.5* for Signal app. Bitwarden Play Store app includes 258 reviews collected for the selected version, while Signal app includes 1,965 total reviews. The high number of reviews collected, will be a strong advantage in seeking external validation from people who are using these apps daily and are invested enough to share their experiences, they can be considered authentic feedback. User interviews and surveys were considered as methods to collect real user feedback but were fast excluded due to time constraints and inability to gather a representative set of users since those apps operate on multiple platforms and regions of the world.

To support the internal validity of UI-related feedback generated by APs, the evaluation framework refers to the *ISO 9241-110:2020* and *ISO 9241-112:2017(E)*.
*ISO 9241-110:2020* is an international standard that defined several principles of user-system interaction such as conformity with user expectation, it guides the development process of UI components to ensure they are intuitive, consistent and aligned with user already established mental models [27]. The *ISO 9241-112:2017(E)* is another international standard guideline that focuses more on how information should be presented, with the end goal of reaching qualities such as clarity and distinguishability [28].
By referencing these guidelines, the study increases confidence that the insights reflect meaningful, human-centered design shortcomings or strengths.

Regarding the reliability of the study, it is important to note that reliability refers to the consistency and the ability of reproducing this study, especially the feedback generated by APs. Although all models are executed under similar condition (same hardware environment, same network settings and prompt structure), some factors may hinder reproducibility. Using nonlocal open-source AI model such as Open AI's GPT-4o that may introduce some changes in the future through updates or infrastructure changes. This might alter the reproduction of the results presented in this study although the screenshots and prompts used with that model are fixed and saved to guaranty some form of consistency throughout time.
On other models such as Meta Llama, Mistral, and DeepSeek the reliability is stronger because those models are downloaded and are running locally on the hardware. This ensure that the same version used in this project can be retrieved and be exploited across tests to obtain the same results.

On the generalization of the results of the study, the artifacts and framework can be applied on similar startup or open-source project. Changing the codebase might result in longer or shorter analysis time for the artifacts. Nevertheless, the core framework remains suitable for teams seeking APs evaluation of their software product as experimentation.

# 4 RESULTS AND ANALYSIS

## 4.1 On the content

The following section presents the results of the APs contextual feedback generated during the demonstration phase. An analysis can also be found, evaluating the relevance of these feedback and the APs' performance, specifically in terms of execution time.

### 4.1.1 Overview of Evaluated Cases

Based on the evaluation of the selected ISO guidlines principles, **6 cases** were selected to review the applications and conduct the assessment of APs feedback (Table 7). Each case focus on a specific task, aspect, or usability challenge related to one of the selected mobile applications (Table 8). To investigate the practicality of APs into realistic application, each case includes a user-facing evaluation performed either by Persona A or B and in some relevant cases a technical code analysis approach by Persona C.

| Case number | Title | Description |
|---|---|---|
| **1** | Password Font Verification | Tests font clarity in password display to distinguish visually similar characters. |
| **2** | *Special Character Display Integrity Check* | Investigates character rendering anomalies when displaying passwords with special symbols. |
| **3** | Android 14 Compatibility | Verifies core functionality behavior and UI performance on Android 14 devices. |
| **4** | Workflow Friction: Photo Attachment | Analyzes the efficiency of photo attachment feature and required interaction steps. |
| **5** | User Interface and UX Modernization | Evaluating interface design elements for contemporary standards and functional accessibility. |
| **6** | UI/UX and Visual Design Limitations | Evaluating the clarity and navigation experience regarding user engagement and cognitive overload. |

Table 7: Evaluation cases description (1)

| | Quality Attribute | Persona Used | App target |
|---|---|---|---|
| **Case 1** | Accessibility, Security | B | Bitwaden |
| **Case 2** | Reliability | B & C | Bitwarden |
| **Case 3** | Compatibility, Reliability | C | Bitwarden |
| **Case 4** | Usability, Efficiency | A | Signal |
| **Case 5** | **Usability, Accessibility** | B | Bitwarden |
| **Case 6** | **Usability, Accessibility** | A | Signal |

Table 8: Evaluation cases description (2)

#### 4.1.1.1 Case 1 Results: Password Font Verification

The test focus on analyzing the readability of the **Bitwarden** displayed passwords by **Persona B**, focusing on the font type and size making it difficult to distinguish specific characters such as digital one and low case L for example (1,2 / low). Users may misread passwords when viewing them, potentially causing authentication failures or incorrect password entries. The case is assessing the accessibility and security of Bitwarden app through computer vision analysis.

**Persona B**

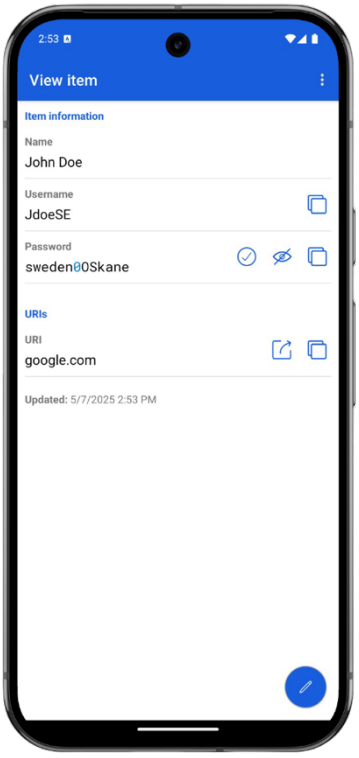
**Prompt**:

*"You are {PersonaB}*

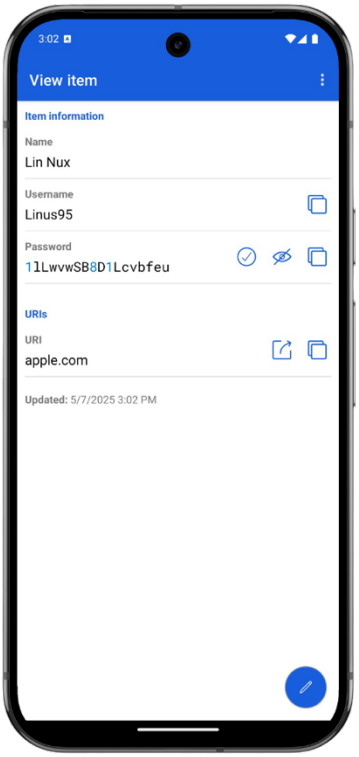
*Please analyze the following UI screenshot with a focus on the readability of the password font in the Bitwarden app.*
*This review supports Case 1: Password Font Verification, which investigates whether the current font choice makes it difficult for users like you to distinguish between visually similar characters — such as "1" (one), "l" (lowercase L), "I" (uppercase i), "0" (zero), and "O" (uppercase o).*
*Your review should answer this case issue by focusing on:*
*- Whether the font type and color make these characters distinguishable*
*- Any confusion when reading or interpreting passwords*
*- Whether the font feels readable and trustworthy under typical usage.*
*Important Instructions:*
*- Only evaluate what is visible in the screenshot, be specific about what you see.*
*- Be specific and detailed*
*- Highlight any flaws or risks that could affect usability, accuracy, or trust.*

*Now analyze this screenshot:"*

**Screenshot provided**:




Password profile of
John Doe - View




Password profile of
Lin Nux - View

**Active Persona Feedback**:

*As a user who values simplicity and clarity, I've generally found the Bitwarden app quite commendable, especially when it comes to password readability and trust. The font used is professional and makes it easy to distinguish commonly confused characters like "o" (lowercase) and "0" (zero), which is important for preventing mistakes. The black font on a white background is a classic choice, offering good contrast and easy readability.*

*However, there are areas where Bitwarden could improve to better accommodate users like me. While the font is clear for most characters, similar ones such as "1" and "l" could be enhanced further for better differentiation. This would be particularly helpful for those with aging eyes or minor visual impairments. An option to adjust font size within the app itself would also be a valuable addition, as many users rely on increased sizes for better readability.*

*Overall, while Bitwarden does a great job in many aspects, refining character clarity and providing more customizable accessibility options would truly elevate the user experience for those of us who prioritize ease of use and trustworthiness.*

**Model used**: GTP-4o
**Elapsed Time**: 39.09 seconds

A more detailed generated evaluation from the Active persona can be found in Appendix 8.1.1

#### 4.1.1.2 Case 2 Results: Special Character Display Integrity Check

The case examine a reported issue in the application Bitwarden , where some specific characters such as "*>@§*<°*" are not appearing accurately in the UI when the user try to view his/ her saved password but remain visible when editing them. The case is assessing the reliability of Bitwarden app through computer vision analysis, Persona B and C are involved in this test.

**Persona B**

**Prompt**:

*"You are {Persona B}*
*Please analyze the following UI screenshot with a focus on the accurate rendering of special characters in the Bitwarden app.*
*This review supports Case 2: Special Character Display Reliability, which investigates a reported issue where certain characters — such as ">@§*<°" — do not appear accurately in the password view mode but remain visible when editing the password.*

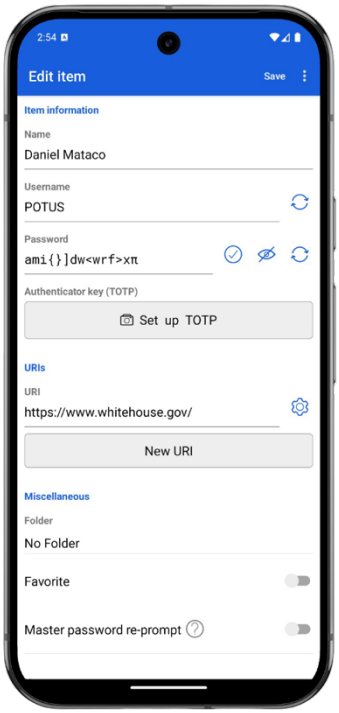
*The purpose of this case is to assess the reliability of the UI's character rendering for users attempting to view their saved passwords.*

*Your review should answer this case issue by focusing on:*
*- reading both password inputs, printing them and compared them and check for any differences between the two*
*- Whether any of the mentioned characters are missing, or visually altered in the screenshot*
*- Any inconsistencies between the view and edit modes in terms of character visibility*
*- Whether the UI feels reliable and consistent for accurately displaying saved passwords*

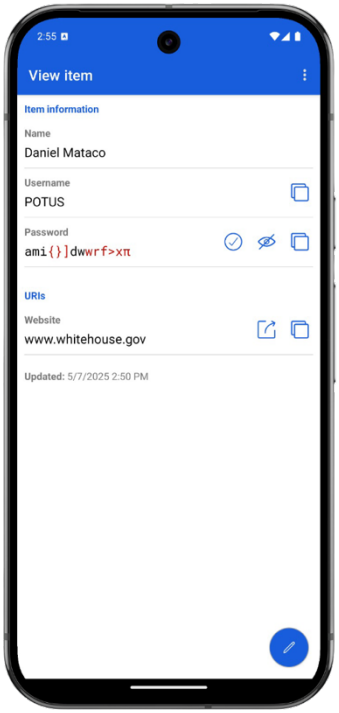
*Important Instructions:*
*- Only evaluate what is visible in the screenshot — be specific and precise*
*- Keep your review detailed but limited to half a page*
*- Highlight any flaws or risks that could affect usability, reliability, accuracy, or user trust*

*Now analyze this screenshot:"*

**Screenshot provided**:




Inputting password with special characters




Viewing the edited password

**Active Persona Feedback**:

In the provided screenshot, comparing the “Edit Item” and “View Item” screens of the Bitwarden app focuses on assessing how special characters are rendered in the password field. The password in the "View Item" screen is displayed as “*ami{}]dw<wrf>xπ*”, containing special characters such as `{}`, `]`, <, and `π`.

However, when switching to the "Edit Item" screen, the password appears as “ami{}]dwwrf>xπ”. Here, it is immediately noticeable that the < character is **missing**. This introduces a discrepancy between what is shown in the view mode and what appears during editing. Such inconsistencies can lead to user confusion or incorrect password use, especially in cases where special characters are required by password policies.

No visible misrendering or alterations for the characters `{`, `}`, `]`, >, or `π` was observed between the two modes. The UI remains visually consistent and user-friendly in terms of layout and structure. However, the missing < character in edit mode is a serious concern, as it directly impacts the **reliability and accuracy** of the password data being displayed and potentially stored.

This inconsistency can undermine user trust in the application, particularly when accuracy is paramount for a secure password manager. Ensuring that all characters are displayed correctly and consistently in both viewing and editing contexts is essential to maintain confidence in the app’s integrity and usability.

**Model used**: GTP-4o
**Elapsed Time**: 32.24 seconds

**Persona C**

1) **Prompt**:

*"You are {PersonaC}. As an expert code reviewer, analyze this code from `{file_path}` evaluate the following case: The case examine a reported issue in the application Bitwarden , where some specific characters such as ">@§*<° " are not appearing accurately in the UI when the user try to view his/ her saved password but remain visible when editing them.*

*Important Instructions: -be concise and accurate*

*-if the file is not related to the objective just mentioned "not related"*

*-include code example (the line in the file or concise citation)*

*CODE:"*

**Active Persona Feedback**:

*To summarize, based on the provided code snippets, it appears that the issue with special characters not appearing accurately in the UI when viewing saved passwords may be related to several factors:*

*1. Handling of non-standard character sets during decoding or encoding processes within the application.*
*2. Correct .NET culture codes being provided for all supported Android languages.*
*3. Exception handling in various functions or methods throughout the application.*
*4. Potential missing support for RTL (Right-to-Left) languages like Hebrew in certain functions or methods.*

***To improve the application and provide feedback, consider**:*

*1. Reviewing and refining the logic for handling non-standard character sets in decoding/encoding processes.*
*2. Ensuring that the correct .NET culture codes are provided for all supported Android languages.*
*3. Streamlining exception handling throughout the codebase to improve readability and maintainability.*
*4. Adding support for RTL languages like Hebrew (`iw`) if it is currently missing in the application.*

***Furthermore, here are some best practices for various code elements**:*

*1. Use clear and descriptive names for views, resources, functions, and classes.*
*2. Organize layouts and code to improve readability and maintainability.*
*3. Define dimensions in a centralized `values/dimens.xml` file instead of using hard-coded values directly within layout files.*
*4. Use appropriate exception handling mechanisms to prevent potential crashes and ensure graceful recovery in case of unexpected conditions.*
*5. Implement consistent naming conventions for elements and resources throughout the codebase.*
*6. Follow Android coding guidelines, such as using Javadoc-style comments and structuring your XML layouts properly.*
*7. Keep lines shorter than 80 characters for better readability and maintainability.*

**Model used**: Mistral v0.3
**Elapsed Time**: 45 minutes, 39.94
**Avg time per file**: 23.72 seconds

2) **Prompt**: Same as 1.

**Active Persona Feedback**:

The issue with special characters (e.g., `>@$*<°`) not displaying correctly in the Bitwarden UI likely stems from **character encoding mismatches**. These characters may be stored correctly but are not rendered properly due to inconsistent or incorrect encoding during display. To fix this, ensure that password strings are consistently handled with Unicode-aware methods—such as using UTF-8 encoding throughout storage, retrieval, and display operations. In Java, for example, the `Charset.forName("UTF-8")` method can ensure proper decoding and rendering of special characters.

In terms of **autofill configuration**, the `autofillservice.xml` file currently lists supported browsers, but improvements can be made by grouping entries more clearly and using consistent naming conventions. This aids maintainability and ensures compatibility with a wider range of browser variants, such as beta or developer versions of Chrome or Edge.

For **UI components** like `EditText`, developers should apply input filters that safely handle special characters while preserving user intent. This prevents misinterpretation or stripping of non-ASCII symbols and ensures accurate password visibility.

Additionally, the method `AndroidToDotnetLanguage` lacks robust **exception handling**. It's recommended to implement consistent `try-catch` blocks—especially for exceptions like `CultureNotFoundException`—to avoid unexpected crashes and to log meaningful error messages for debugging.

By addressing these concerns—encoding consistency, configuration clarity, UI rendering accuracy, and proper exception handling—the application can offer a more reliable and user-friendly experience, particularly for users relying on special characters in their credentials.

**Model used**: Llama3.2
**Elapsed Time**: 34.4 minutes
**Avg time per file**: 26.03 seconds

3) **Prompt**: Same as 1.

**Active Persona Feedback**:

Okay, so I'm trying to figure out why some specific characters aren't showing up correctly when displaying saved passwords in the Bitwarden app on Android. The user mentioned that characters like "=@§*<°" don't appear accurately. Hmm, let me think about how this could happen.

First, maybe it's related to how the app decodes or encodes these special characters before they're displayed. I remember that sometimes when you save passwords in an app, especially on mobile devices, there can be issues with encoding/decoding processes due to different locales or character sets being used between the server and the device.

I should check if Bitwarden is handling Unicode properly. Maybe it's using UTF-8 consistently across all platforms. But wait, some characters might require specific encodings depending on the locale settings of the device. If the app doesn't account for these locale-specific encodings when saving or retrieving passwords, those special characters could get lost or misencoded.

Another angle is how Bitwarden interacts with its local storage or databases to store these passwords. Perhaps there's an issue in how it's encoding the data before storing it and decoding it upon retrieval. For instance, using incorrect MIME types for attachments might corrupt non-ASCII text when saved to a file.

Also, I should consider the display settings on the device itself. If the password string is being displayed as plain text without proper font handling or if there are missing characters in certain fonts, that could lead to special characters not showing correctly.

Maybe it's also related to how Bitwarden interacts with third-party services for password management. If those services don't support extended Unicode characters, they might truncate or drop some characters when the password is displayed back on the device.

I think the most likely culprit is improper handling during storage or retrieval of these passwords within the Bitwarden app itself. Maybe the encoding/decoding process isn't correctly preserving certain characters, especially those that require specific locale-specific encodings.

Additionally, if Bitwarden's localization files (LC) are misconfigured for certain languages that use these special characters, this could also lead to them not displaying correctly when viewed in the UI.

Another possibility is related to how Bitwarden reads from or writes to system fonts. If it's relying on system fonts that don't properly support all Unicode points, those special characters might not render correctly either.

**Model used**: DeepSeek R1
**Elapsed Time**: 56.07 minutes
**Avg time per file**:45.13 seconds

#### 4.1.1.3 Case 3 Results: Android 14 Compatibility

This case investigates the compatibility of **Bitwarden** with Android 14. Animation stuttering and functionality failures were identified as issues on the latest OS version. This case use **Persona C** and operate a codebase analysis to trace the potential cause of these issues.

**Persona C**

1) **Prompt**:

> *"You are {Persona C}. As an expert code reviewer, analyze this code from {file_path} to evaluate the following case: This case investigates the compatibility of Bitwarden with Android 14. Animation stuttering and functionality failures were identified as issues on the latest OS version. Operate a codebase analysis to trace the potential cause of these issues.*
>
> *Important Instructions:*
>
> *- If the file is not related to the objective, skip it and don't write anything*
>
> *-only focus on the issue*
>
> *- Include code examples (the line in the file or a concise citation) that supports your analysis.*
>
> *CODE:"*

**Active Persona Feedback**:

> Based on the provided code analysis, I will provide a suggested refactoring plan to address potential compatibility issues with Android 14: Refactoring Plan:
>
> **1. Remove unnecessary nesting:**
> • Simplify the XML structure by removing unnecessary nested elements.
> • Use a consistent naming convention for all package elements.
>
> **2. Apply consistent naming conventions:**
> • Ensure that all package names follow a consistent format (e.g., lowercase with hyphens Instead of underscores).
> • Update existing code to reflect the new naming convention.
>
> 3. **Ensure all packages have** *android:supportsInlineSuggestions* **attribute set to true:**
> • Review the XML file and update any packages that do not have this attribute.
> • Ensure that all packages are consistent in their behavior regarding inline suggestions.
>
> **4. Optimize XML size (if necessary):**
> • If the XML file is too large or complex, consider using compression techniques or optimizing its structure to reduce its size.
>
> **5. Add clear and concise comments:**
> • Provide detailed comments explaining the purpose of each package and configuration option.
> • Use clear and descriptive variable names throughout the code. 6. Review and test refactored code:
> • Verify that the refactored code meets all compatibility requirements for Android 14.
> • Test the code thoroughly to ensure it works as expected.
>
> **By following this refactoring plan, you can improve the maintainability and readability of your XML code while addressing potential compatibility issues with Android 14.**

**Model used**: Llama3.2
**Elapsed Time**: 26 minutes
**Avg time per file**: 14.05 seconds

#### 4.1.1.4 Case 4 Results: Workflow Friction: Photo Attachment

This case provides feedback on the number of steps required to join a picture on the **Signal** app and send it to a friend. This is a popular use case in the app, if too heavy, it can lead to frustration and slower communication for users who frequently share media. **Persona A** is used to assess the usability and efficiency of this case. (Computer vision evaluation)

**Persona A**

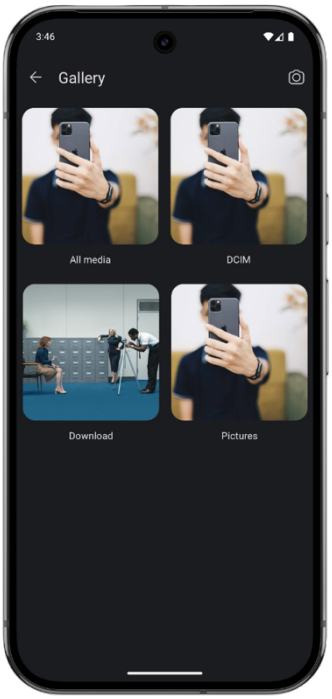
**Prompt**:

> *"You are {Persona A}*
>
> *Please analyze the following UI screenshot with a focus on the number of steps required to attach and send a picture to a contact in the Signal app.*
>
> *This review supports Case 4: Media Sharing Usability, which evaluates whether the process of selecting and sending a picture to a friend involves too many steps. This is a common and important use case in Signal, and if the flow feels too long or cumbersome, it may lead to user frustration and slower communication.*
>
> *Your review should answer this case issue by focusing on:*
> *- How intuitive and efficient the photo-sharing process appears based on the visible interface*
> *- Whether the steps seem excessive or unnecessary*
>
> *Important Instructions:*
> *- Only evaluate what is visible in the screenshot — be specific and precise*
> *-focus on the number of steps, you can use the file name as indicator, your answer should be focus around it*
>
> *Now analyze this screenshot:"*

**Screenshot provided**:




Step 0-Initial conversation view




Step 2 - Gallery Selector




Step 5 – Photo Validation

6 Screenshots in total were provided to the Active Persona for this case, they can found in Appendix 8.2.

**Active Persona Feedback**:

Based on the provided screenshots, the process of attaching and sending a picture in the Signal app involves the following steps:

**Initial Conversation View (Step 0)**: Start from the conversation window. This is intuitive as it provides immediate access to communication tools.

**Camera App (Step 1)**: This step allows taking a new photo directly, which is straightforward for capturing real-time images. However, users seeking to share existing photos have to navigate further.

**Gallery Selector (Step 2)**: Redirects to the gallery overview, showing different album options. This can feel like an extra step if users primarily access a specific folder frequently.

**Folder Selection (Step 3)**: Navigate to a specific folder. While organized, it might feel unnecessary if frequent photos are in the same location.

**Photo Selection (Step 4)**: Users select the desired photo. This step is essential for accuracy in choosing the right image.

**Photo Validation/Edition (Step 5)**: Provides options to edit or confirm the image before sending. This is useful but could be optional for users who prefer quicker sends.

**Photo Sent Back to Initial View (Step 6)**: Completes the process by returning to the chat with the photo sent. This closure is efficient, providing confirmation the action is completed.

**Analysis**:

**Intuitiveness and Efficiency**: The flow is generally straightforward but might feel slightly cumbersome due to the multi-step process, particularly Steps 2 and 3. These could be streamlined by remembering frequent folders or integrating more directly with the chat window.

**Excessive Steps**: Steps 2 and 3 could potentially be merged or optimized to reduce friction. If a user often sends pictures from a specific album, customizing shortcuts could improve efficiency.

**Model used**: GTP-4o
**Elapsed Time**: 2 minutes and 3 seconds

#### 4.1.1.5 Case 5 Results: User Interface and UX Modernization

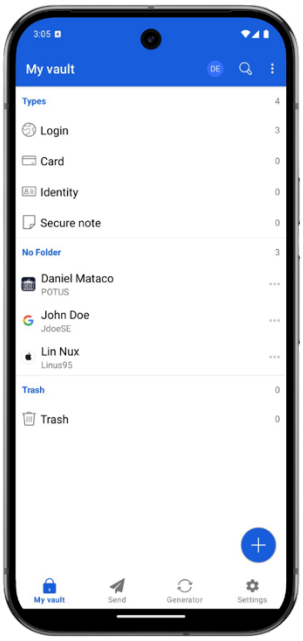
This Case evaluate the overall quality of **Bitwarden** Ui and UX component and visual appeal, **Persona B** will evaluate the usability and accessibility of the app, by judging on whether the UI elements are outdated or relevant to users' needs through computer vision.

**Persona B**

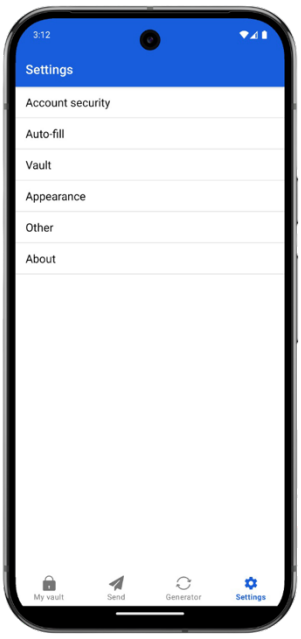
**Prompt**:

*"You are {Persona B}*
*Please analyze the following UI screenshot with a focus on the overall quality of the Bitwarden app's UI and UX components.*
*This review supports Case 5: UI and UX Evaluation, which assesses the visual appeal, usability, and accessibility of Bitwarden's interface. The goal is to determine whether the UI elements appear outdated or aligned with current user needs, using visual cues visible in the screenshot.*
*Your review should answer this case issue by focusing on:*
*- Whether the UI elements (buttons, layouts, icons, spacing, etc.) feel modern, intuitive, and accessible*
*- Any signs of outdated design patterns or components that could hinder user experience*
*Important Instructions:*
*- Only evaluate what is visible in the screenshot — be specific and precise*
*- Focus your answer on the UI elements shown in the file, and use the file name as an indicator if helpful*
*Now analyze this screenshot:"*

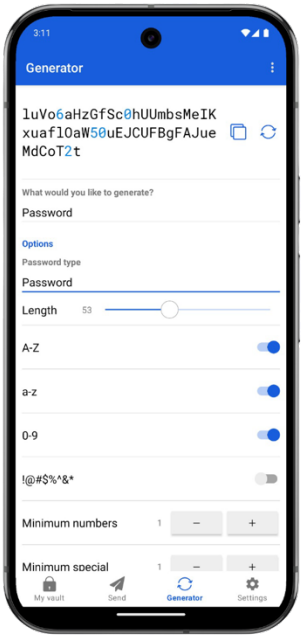
**Screenshot provided**:




Main View of the app
My vault




Settings View




Password Generator View

5 Screenshots in total were provided to the Active Persona for this case, they can found in Appendix 8.3.

**Active Persona Feedback**:

"As a daily user who highly values simplicity and clarity, I find the Bitwarden app to be mostly intuitive and accessible. The user interface, particularly in the 'My Vault' and 'New Send' views, is clean and uncluttered, which helps me navigate and complete tasks without feeling overwhelmed. The use of clear typography, ample white space, and recognizable icons is excellent for readability and ease of use, especially for someone like me who adjusts font sizes for accessibility

The consistent navigation elements and minimalistic design across different sections, such as 'Settings' and 'Generator,' align well with my needs for straightforward interaction. However, the visibility of some interactive elements, like the three-dot menus and 'Options' dropdown, could be improved by making them more prominent with color cues or added icons. This would help in more smoothly identifying actionable items without hesitation.

Though the design generally avoids technical jargon, the occasional use of terms like special characters could be a bit challenging. Providing tooltips or help icons would be beneficial to help less tech-savvy users understand functions more intuitively.

Overall, the Bitwarden app successfully balances modern design with accessibility, making it a trustworthy companion for managing my digital tasks with ease. Enhancements in interactive elements' visibility and some additional user guidance could further improve the experience for aging yet active users like me."

A more detailed generated evaluation from the Active persona can be found in Appendix 8.1.2

**Model used**: GTP-4o
**Elapsed Time**: 1.30 minutes

#### 4.1.1.6 Case 6 Results: UI/UX and Visual Design Limitations

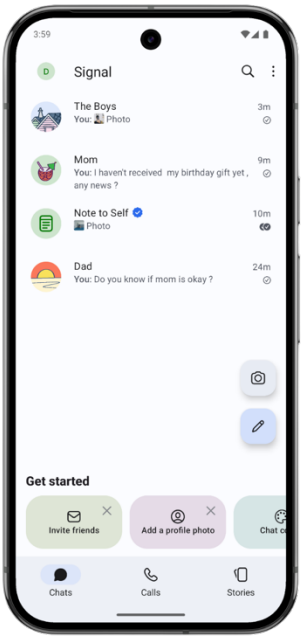
**Persona A** will evaluate the overall User Interface of **Signal** app and provide insights on the modernity of the UI elements. Evaluating if it is visually attractive and focus specifically on the distracting design elements of the app. Providing insights on the usability and accessibility of the Signal App through computer vision.

**Persona A**

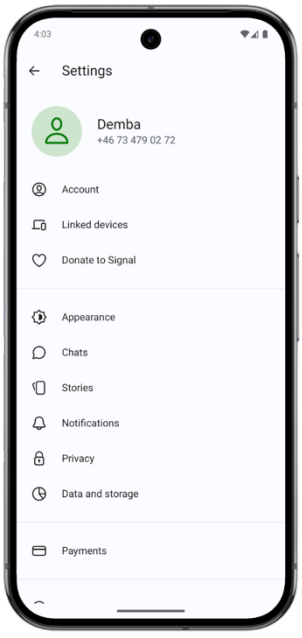
**Prompt**:

> *"You are {Persona A}*
>
> *Please analyze the following UI screenshot with a focus on the overall quality of the Signal app's User Interface (UI). This review supports Case 6: UI Evaluation for Signal, which assesses the modernity, visual appeal, and accessibility of Signal's interface.*
> *The goal is to evaluate whether the UI elements are visually attractive and if any distracting design elements hinder usability and accessibility.*
> *Your review should answer this case issue by focusing on:*
> *- Whether the UI elements (buttons, layouts, icons, spacing, etc.) feel modern, intuitive, and accessible*
> *- Any signs of outdated design patterns or components that could hinder user experience*
> *Important Instructions:*
> *- Only evaluate what is visible in the screenshot — be specific and precise*
> *- Focus your answer on the UI elements shown in the file, and use the file name as an indicator if helpful*
> *Now analyze this screenshot:"*

**Screenshot provided**:




Main View of the app – Chats




Settings View




Call View

5 Screenshots in total were provided to the Active Persona for this case, they can found in Appendix 8.4.

**Active Persona Feedback**:

"As a tech-savvy user immersed in digital ecosystems, I find the Signal app's UI to be mostly aligned with modern design standards, offering a clean and intuitive experience across various screens. The app's layout generally maintains ample spacing and minimal distractions, which is a big plus for someone who values efficiency and functional design. Each key element, from the search bar to intuitive icons for Chats, Calls, and Stories, is well-placed, supporting fast interaction without overwhelming the user.

While the overall aesthetics are appealing, there are minor areas for refinement. The "Get started" section could benefit from more contemporary styling with updated colors and shapes, better aligning with the sleekness of the rest of the app. Also, the camera and edit buttons could blend more seamlessly into the design to avoid disrupting the otherwise pleasant visual flow.

Additionally, while the calls view is clear and uses whitespace effectively, slight adjustments in color contrast could further enhance visibility, especially on smaller screens. Accessibility improvements such as larger icons and text would cater to a wider range of users.

The group info view is particularly strong in organization, featuring friendly, rounded icons and sections for key settings like "Disappearing messages" and "Chat color & wallpaper." My only suggestion here would be to incorporate more customization options directly on this screen to enhance user experience further.

In the settings view, the well-organized hierarchy and minimalist design ensure a smooth navigation experience. However, a touch of dynamic elements or subtle color highlights could make interactions even more engaging.

Finally, the story guidance screen feels modern and internationally comprehensible, with easy-to-see icons and clear, accessible text. Ensuring familiarity with gesture-based navigation would further bolster user accessibility.

In summary, the Signal app largely succeeds in offering a seamless and stylish user experience that meets the expectations of fast-adapting Gen Z users like myself. With a few tweaks, it could further solidify its standing as a leading app in efficient, feature-focused design."

**Model used**: GTP-4o
**Elapsed Time**: 1.41 minutes

## 4.2 Execution Time Analysis

After evaluating each case, the following running time were recorded for the active persona feedback analysis during the evaluation. These results testify on the APs abilities to generate diverse type of user feedback with efficiency while processing varying volume of inputs and data across cases.

| Model | Average time |
|---|---|
| GPT- 4o | 1 minutes and 14 seconds |
| Llama3.2 | 30 minutes and 12 seconds |
| Mistral | 45 minutes and 39 seconds |
| DeepSeek-R1 | 56 minutes amd 7 seconds |

*Table 9 Average running time by model*

GPT-4o has the fastest execution time when it comes to generate feedback from images ranging from 2 to 6 pictures (Table 9). When it comes to codebase analysis Llama presents the fastest execution time, with 30 minutes and 12 seconds to analyze the full codebase and generate feedback based on a specific case. As opposed to DeepSeek-R1 that has the longest execution time with 56 minutes and 7 seconds. A good compromise between those two models is Mistral, the model analyzed the full code base and provided its evaluation in 45 minutes and 39 seconds.

## 4.3 ISO Alignment and User Review Comparison

The feedback generated by the different APs, turned out to be valuable, aligning well with a ISO guidelines and reflects real user concerns, assessment and practical issues. Find below an assessment of this alignment:

Case 1: Password Font Verification

<table>
<tr><th colspan="2">Consistency</th></tr>
<tr><td colspan="2">Guidance related to using presentation attributes:<br><br>1) 6.3.3.1 Items of information that are logically dissimilar should be presented differently (e.g. using different presentation attributes) to make their differences obvious.<br><br>NOTE: This can involve locations, visual shapes, sizes, colours, typefaces, etc.. ; auditory order of presentation, volumes, tones, frequencies, etc ..; tactile locations, shapes, vibrations, pressures, etc.</td></tr>
<tr><td>APs feedback</td><td>User Feedback</td></tr>
<tr><td>As a user who values simplicity and clarity, I've generally found the Bitwarden app quite commendable, especially when it comes to password readability and trust. The font used is professional and makes it easy to distinguish commonly confused characters like "o" (lowercase) and "0" (zero), which is important for preventing mistakes. The black font on a white background is a classic choice, offering good contrast and easy readability.<br>However, there are areas where Bitwarden could improve to better accommodate users like me. While the font is clear for most characters, similar ones such as "1" and "l" could be enhanced further for better differentiation. This would be particularly helpful for those with aging eyes or minor visual impairments. An option to adjust font size within the app itself would also be a valuable addition, as many users rely on increased sizes for better readability….</td><td>• Review #230: "Generally very good UI for Bitwarden. One caveat - please use font that distinguishes between 'O' and '0' (capital O and zero) for shown passwords."</td></tr>
</table>

Table 10: Consistency Principle Case 1

In this case, the AP praise the UI and strongly aligns with ISO guidance, especially with section 6.3.3.1 by addressing confusing character like zero and letter o and other similar issues. The feedback goes even further by analyzing the lack of font size customization and accessibility features. The feedback also connects with issues that user encountered (feedback #230) but complete it by suggesting enhancement.

Case 2: *Special Character Display Integrity Check*

<table>
<tr><th colspan="2">Consistency</th></tr>
<tr><td colspan="2">Guidance related to structuring of presented information<br><br>2) 6.6.2.3 Where attributes are used to encode properties of pieces of information, they should be used consistently throughout the interactive system, or at least throughout a single state of the interactive system.<br><br>NOTE: Where encoding attributes are used differently in different states of an interactive system, it is important that changes to the state of the interactive system that also change the encoding are made clear to the user.</td></tr>
<tr><td>AP feedback</td><td>User Feedback</td></tr>
<tr><td>In the provided screenshot, comparing the “Edit Item” and “View Item” screens of the Bitwarden app focuses on assessing how special characters are rendered in the password field. The password in the "View Item" screen is displayed as “ami{}]dw<wrf>xπ”, containing special characters such as {}, ], <, and ⊓.<br><br>However, when switching to the "Edit Item" screen, the password appears as “ami{}]dwwrf>xπ”. Here, it is immediately noticeable that the < character is missing. This introduces a discrepancy between what is shown in the view mode and what appears during editing. Such inconsistencies can lead to user confusion or incorrect password use, especially in cases where special characters are required by password policies. No visible misrendering or alterations for the characters {, }, ], >, or ⊓ was observed between the two modes. The UI remains visually consistent and user-friendly in terms of layout and structure. However, the missing < character in edit mode is a serious concern, as it directly impacts the reliability and accuracy of the password data being displayed and potentially stored.</td><td>• Review #26: "A solid app, but I did find a bug. If you have a password with a less-than symbol (i.e. <) in it, and you click on the eyeball to display the password, the string that it displays will be missing the less-than symbol. If you edit the password, you will see the less-than symbol. Please fix! Thank you!"</td></tr>
</table>

Table 10: Consistency Principle Case 2

In this case, the AP managed to identify a critical UI inconsistency which strongly aligns with section 6.6.2.3. The AP feedback is contextual and accurate about the misrepresentation of one character. The AP value and depth to real user feedback, Review #26 identified this issue, while the comment is concise and informative, the AP feedback is more elaborate and explore others implication such as user trust, and reliability.

Case 3: Android 14 Compatibility

| **Controllability** | |
|---|---|
| The interactive system allows the user to maintain control of the user interface and the interactions, including the speed and sequence and individualization of the user-system interaction. | |
| AP feedback | User Feedback |
| Based on the provided code analysis, I will provide a suggested refactoring plan to address potential compatibility issues with Android 14: Refactoring Plan:<br>…<br>**By following this refactoring plan, you can improve the maintainability and readability of your XML code while addressing potential compatibility issues with Android 14.** | • Review #26: "A solid app, but I did find a bug. Review #242: "In Android 14 the automatic filling function cannot be activated I hope this problem will be solved in the next update"<br>• Review #66: "The animation is very stuttering when I open a login to edit, please help to fix (Galaxy A15 with Android 14)" |

Table 11: Controllability Principle Case 3

In this case, the AP took a code analysis approach to find potential clues in what might influence the app to crash under OS version 14, the AP feedback provide a refactoring plan and doesn't really match any conclusion on whether Android 14 present issues. The feedback generated does really align to the principal of controllability since it only provides guidelines. On the other end the user feedback are detailed and specific about which aspects of the app are not working properly, therefore not controllable.

Case 4: Workflow Friction: Photo Attachment

| **Conciseness** | |
|---|---|
| **Guidance related to conciseness of content**<br>**6.5.2.1** Minimalism and simplicity should be used to present information concisely. | |
| AP feedback | User Feedback |
| **…Analysis**:<br>**Intuitiveness and Efficiency**: The flow is generally straightforward but might feel slightly cumbersome due to the multi-step process, particularly Steps 2 and 3. These could be streamlined by remembering frequent folders or integrating more directly with the chat window.<br>**Excessive Steps**: Steps 2 and 3 could potentially be merged or optimized to reduce friction. If a user often sends pictures from a specific album, customizing shortcuts could improve efficiency. | • Review #1688: *"Messaging is easy. Attaching photos is annoyingly complicated.... Too many steps & confirmations and then send a photo."*<br>• Review #1681: *"Good basic messaging app, photo upload a bit clunky."* |

Table 12: Conciseness Principle Case 4

The AP generated feedback aligns well with the principle of conciseness by providing insight on simplicity and reducing the number of steps required. The AP provide a more detailed breakdown compared to user feedback. The feedback even suggest improvement that are valid and in line with the sentiment that user express. Review 1688 and 1681 are drawing the same conclusion than the findings of the AP feedback.

Case 5: User Interface and UX Modernization

| **User Engagement** | |
|---|---|
| The interactive system presents functions and information in an inviting and motivating manner supporting continued interaction with the system. | |
| AP feedback | User Feedback |
| "As a daily user who highly values simplicity and clarity, I find the Bitwarden app to be mostly intuitive and accessible. The user interface, particularly in the 'My Vault' and 'New Send' views, is clean and uncluttered, which helps me navigate and complete tasks without feeling overwhelmed. The use of clear typography, ample white space, and recognizable icons is excellent for readability and ease of use, especially for someone like me who adjusts font sizes for accessibility<br><br>The consistent navigation elements and minimalistic design across different sections, such as 'Settings' and 'Generator,' align well with my needs for straightforward interaction. However, the visibility of some interactive elements, like the three-dot menus and 'Options' dropdown, could be improved by making them more prominent with color cues or added icons. This would help in more smoothly identifying actionable items without hesitation.<br><br>Though the design generally avoids technical jargon, the occasional use of terms like special characters could be a bit challenging. Providing tooltips or help icons would be beneficial to help less tech-savvy users understand functions more intuitively.<br><br>Overall, the Bitwarden app successfully balances modern design with accessibility, making it a trustworthy companion for managing my digital tasks with ease. Enhancements in interactive elements' visibility and some additional user guidance could further improve the experience for aging yet active users like me." | • Review #64: "A good password manager that can be used on a selfhosted server. The only improvement I expect is the app UI/UX which is very much outdated and looks like an android 4 app. Hopefully the devs can improve the overall appearance of the app to look more modern."<br><br>• Review #233: "The app idea and security are great! But UX is very bad, and miss some crucial functions. For example, why I can't copy to clipboard the cardholder name? Or any field value from an identity record? Please involve UX designers into developing!"<br><br>• Review #96: "...BUT THE NEW WEBSITE/BROWSER UI SUCKS!... I HATE THE 12/2024 UI UPDATE." |

Table 13: User Engagement Principle Case 5

The AP generated feedback aligns well with the principle of user engagement, it praises UI elements and the consistence of the app. The AP feedback is rich and praise many aspects of the UI and UX, it even promotes improvements and recommendations. On the other hand, much real user feedback does not agree that the app follow these principles, all lot of feedback are composed of just the word “good

UI" or "bad UI" review 64, 233, and 96 are considered representative of user not liking the UI/UX of the app.

Case 6: UI/UX and Visual Design Limitations

<table>
<tr><th colspan="2">Freedom Of Distraction</th></tr>
<tr><td colspan="2">Guidance on minimizing and avoiding distractions<br><br>6.2.1 Principle of freedom from distraction<br><br>Presented information is free from distractions if the information is presented so that required information [1] will be perceived without other presented information interfering with its perception. Distractions from a user's point of view can result [2] from both distracting events and information overload.<br><br>Freedom from distraction involves guidance related to<br>a) avoiding distractions, and b) minimizing distractions.</td></tr>
<tr><td>AP feedback</td><td>User Feedback</td></tr>
<tr><td>"As a tech-savvy user immersed in digital ecosystems, I find the Signal app's UI to be mostly aligned with modern design standards, offering a clean and intuitive experience across various screens. The app's layout generally maintains ample spacing and minimal distractions, which is a big plus for someone who values efficiency and functional design. Each key element, from the search bar to intuitive icons for Chats, Calls, and Stories, is well-placed, supporting fast interaction without overwhelming the user.<br><br>While the overall aesthetics are appealing, there are minor areas for refinement. The "Get started" section could benefit from more contemporary styling with updated colors and shapes, better aligning with the sleekness of the rest of the app. Also, the camera and edit buttons could blend more seamlessly into the design to avoid disrupting the otherwise pleasant visual flow…</td><td>• Review #941: "The app is good, easy to use and does what's intended. It has annoying popups and banners etc, so not enjoying that part. A messaging app needs a clean simple interface with no distractions."<br><br>• Review #953: "The app layout and design could've been better, but still it's smooth, gets things done and is easy to connect with each other."<br><br>• Review #1469: "I do not find this app as intuitive to use or visually attractive as it could and should be. Wanted to switch to it as my primary communication app but can't make it past one text thread."<br><br>• Review #421: "I like this app! It's a little clunky and confusing at first, and not entirely user friendly yet cuz it's so small, but I still think it's helpful to be sure that your msgs are private."</td></tr>
</table>

Table 14: Freedom Of Distraction Principle Case 6

The AP feedback for case 6 generated comprehensive feedback that aligns well the guidelines of freedom of distraction, the AP evaluate the UI of signal and identifies key strength on how information is presented and the lack of distraction on the interface. The feedback assesses the conformity of the app to the guidelines while some real users feedback is not praising those aspects, they often disagree on the simplicity of the app and the distracting event that might occurred.

# 5 DISCUSSION

## 5.1 Interpretation of Feedback Quality

### 5.1.1 Active Persona Powered by GTP-4o

The results from the AP powered by GPT-4o presents undeniable qualities in contextualization and interpretability. On the 5 cases introduces for this model, the Active Persona was able to provide contextual feedback and accurately detect issues, assess the strength of the user interface on both applications. In all cases the AI provided deep analysis comparable to real user feedback, it even surpassed them by providing actionable actions and recommendation for better usability. When compared to real user feedback, APs feedback is more structured, detailed and use a vocabulary that is appropriate, rich and straightforward. The screenshot provided were analyzed very well by the model, in Case 2 the AP could read the passwords from the view and contextually use them in the evaluation to detect unmatching characters and solve the case issue.

The feedback generated by APs were based on the implementation of two Persona A and B, in some feedback (e.g. Case 2) the answer is fully shape around persona B character and provide insight on accessibility and how to make the UI more accommodating for this type of users, such as "*This would be particularly helpful for those with aging eyes or minor visual impairments. An option to adjust font size within the app itself would also be a valuable addition, as many users rely on increased sizes for better readability …"*.
These recommendations would have not been generated if persona A was used in this case, the influence that Personas had on most the feedback generated is noticeable and provided a more authenticity to the feedback. Few of them impersonating user by providing the feedback more humanely, for example Case 6 started as "As a tech-savvy user immersed in digital ecosystems, I find the Signal app's UI to be" and Case 5 started as "As a daily user who highly values simplicity and clarity, I find the Bitwarden app to be". The use of the first person and the accuracy of the feedback is a real strength and blurs the line even further on the distinction between human and APs generated feedback. Although some rigidity in the vocabulary can be observed in most feedback especially with transitions words as "In summary", "However", "Finally", "Additionally,", "Overall". Those words are used very mechanically in most feedback and make the evaluation less genuine.
In some 2 cases (Case 2 and 4), although the personas A and B were used, the feedback generated resulted in very procedural feedback, erasing totally the persona to focus on the case issue uniquely. The answers provided a straightforward approach to the issue by focusing on identifying the issues, analyzing the screenshot provided, and generating a raw evaluation. The evaluations are still valuable and exploitable, but it raises serious interrogation on whether the AP took into consideration the persona or not, and if it shapes the feedback as that persona. As there are no specific tools to know what's happening during the processing by the model, it is a vulnerability that practitioners cannot ignore.

As demonstrated in the results section, the ISO usability guidelines selected served as validation and were used to assess the quality of the APs feedback. All the feedback generated with GPT-4o aligns with some of those guidelines and are even superior to real user feedback. Superior in a sense that AP feedback was able to identify issues or qualities that real users identified, but AP went even further by complementing their feedback with insights and recommendations. One key difference between AP generated feedback and real user feedback is that most reviews written by real users are based on emotion. These reviews are often short, subjective and focus on frustration or satisfaction, they cannot be considered constructive. In contrast AP feedback are structured consistent and might offer deeper value for developers.

### 5.1.2 Active Persona Powered by Mistral, Llama, DeepSeek

The feedback generated with the APs that is using Persona C the code analysis expert turned out to be underwhelming in practice for difference reasons. The AP with GPT-4o was based on computer vision and text base, with an average running time of 1 minutes and 14 seconds that was greatly influences by the number of pictures provided, the more picture provided resulted in higher execution time. Mistral, Llama, and DeepSeek are text-based models and did not support images. It's therefore a codebase analysis approach to try to solve the cases. The time for analyzing each codebase and provide feedback was 30 minutes and 12 seconds for Llama3.2 (average), 45 minutes and 39 seconds for Mistral, and 56 minutes and 7 seconds for DeepSeek-R1. Compared to the execution time of GTP-4o this is a more heavy and less flexible approach. It was also demanding on the components of the machine performing those evaluation.

Regarding the quality of the feedback, the AP powered with Llama provided the most contextual and qualitative feedback. *In Case 2*, where the password on view mode was missing a character, Llama was the only in the 3 models to generate feedback that targeted files related to the case issues and provided worth exploring solution to solve the issues, such as located exception handling.
Mistral and DeepSeek evaluation, relied on providing checklists and generic instruction such as "*Follow Android coding guidelines, such as using Javadoc-style comments and structuring your XML layouts properly.*", that are very generic and not specific enough to solve the issue. Behind these lengthy checklists, there is an inability for the APs to identify the issue through the code base and provide a valuable evaluation. DeepSeek use very hypothetical assumption with word such as "Maybe", "I should", "Perhaps", which decreased the confidence in using the feedback as a reliable source.

Regarding the alignment of the generated feedback with ISO guidelines, since it takes a codebase analysis approach, most of the outcomes apply for software practitioners on how to solve an issue observed in the app. The theme of the review is partially aligned with ISO principles, in *Case 3* for example, the AP feedback provide a guideline to fix the issue with the end-goal of reaching the principle of controllability. It is hard to assess the quality of that feedback since it requires additional investigations with the codebase.

These results highlight serious limitations in using APs as a standalone tool for codebase evaluation, a combination of the two APs (GPT / Llama) may offer a more balanced assessment for software developer, although Llama demonstrated interesting capabilities in analyzing the codebase. For front facing issues involving end-users, GPT proved to be more effective in interpreting visual elements and simulate user interactions. The feedback generated by the APs (Persona C) might be a good start for investigating issues related to user, but it won't provide a deep evaluation and will struggle to identify the root cause of the issue.

In reaction to the underwhelming quality of the feedback provided by the models, it has been considered to modify generation parameters of each model, which might lead to better outcomes. Parameters such as *temperature*, *top_p*, and *repeat_penalty.* They were adjusted from the Ollama command line to explore whether changes in randomness, sampling diversity, and repetition could enhance the quality of the generated feedback. While these adjustments resulted mainly in stylistic variations in tone and style of writing and slightly better running time execution —they did not result in meaningful improvement regarding the technical accuracy and contextuality of the feedback. The outcomes suggest that parameter tuning is effective in how a model expresses itself and how it can run slightly faster at execution, it is not sufficient by itself to overcome the limitations of the models in performing in-depth codebase analysis.

A potential enhancement would be to integrate specialized statistical analysis tools such as SonarQube as a complementary tool that will provide a more detailed technical evaluation. Future research could explore hybrid approaches, where APs generate accurate feedback and SonarQube can validate them by comparing the two analyses. Including this layer could improve both the depth and practical applicability of feedback, especially regarding identifying performance issues, code smells and logic flaws.

### 5.1.3 The importance of prompt engineering

Another aspect that was key in conducting the evaluation of Bitwarden and Signal app was the designing of prompt sent to the APs. Prompts were revised multiple times to achieve acceptable results, it has been noticed that the prompt play a major role in the quality and relevance of the generated feedback. The results section of this report includes prompts that were designed to suit each specific case, depending on the issue selected the prompt was adjusted to better investigate the issue and instructed on what element to focus. In *Case 4* the case was to investigate the number of steps required to upload a photo and send it on Signal, the prompt used had to be very descriptive about what the AP should focus on. "*How intuitive and efficient the photo-sharing process appears based on the visible interface, Whether the steps seem excessive or unnecessary, focus on the number of steps, you can use the file name as indicator, your answer should be focus around it".*

Without those instructions the APs will just analyze each picture as a standalone case and will not understand that they are connected and are part of a workflow. The results were greatly impacted by the quality of the prompt. It was an expected outcomes that was outlined in the related work section of the report. In section 3.1.14, the authors in their findings pointed out that unfocused or poorly designed prompt often led to misalignment and irrelevant outputs. The findings of this experiment clearly reflect their observations. It has been emphasized that the clarity and structure of a prompt directly affect how large language models simulate user behavior and provide accurate contextual feedback [19]. The findings in the article [19] are particularly relevant in the case of persona-guided UI/UX evaluations, where vague prompts introduce limitations in the model abilities to understand user goals and behaviors. Similarly, a related study reported that insufficiently detailed prompts often resulted in generic, inaccurate, and sometimes even hallucinated responses when applying generative AI in adaptive UX scenarios [20]. In the finding of our study, APs that received structured prompts by explicitly defining the persona, task, and intended workflow generated usability feedback that was more aligned with ISO standards and more consistent when comparing them with real user concerns. These results confirm that prompt engineering is not only essential to accurate generation but also foundational in achieving meaningful feedback through APs.

Another point to mention is the selection of the cases, the cases were selected based on issues or strength that were noticed navigating on both apps, but their evaluation was guided in the prompt for APs to generate relevant feedback. To the exception of *Case 5* and *Case 6* where the overall user interface was analyzed and less guidance was required, the others case required more specific guidance. In *Case 2,* its clearly indicated how to fix the issue in the prompt *"reading both password inputs, printing them and compared them and check for any differences between the two".* Without such instruction the AP wouldn't be able to detect the issue on the screenshot provided and the feedback would have focus on usability resulting in more descriptive analysis. As also stated in the related work 3.1.14, the necessity of having human supervision is essential when generating APs feedback. It is a key element in having generated feedback that are qualitative, contextual and actionable.

## 5.2 Usefulness in a Startup Context

The APs feedback generated with Persona A and B are promising and can be considered valuable requirements engineering artifacts. The use of GTP-4 as a model that can analyzed text and images was a key element in making APs efficient and relevant. All the feedback generated this AP are align with official ISO guidelines on usability principles and user-system interaction, this alignment validates the concept of AP generated feedback and their relevance into evaluating software. But more importantly the generated feedback was in adequation with real user feedback, for startup the APs can represent an innovative solution to evaluate their software product with only a few resources.
The alignment between AP-generated feedback and real user concerns is consistent with previous research that demonstrated how large language models (LLMs) can replicate user feedback in adaptive UX evaluation scenarios [19]. As opposed to that study, which is primarily exploring synthesized personas and textual analysis, our present work integrates multimodal inputs (screenshots and text) and ISO-based usability principles to assess contextual alignment which expands the applicability of LLMs in early stage design and validation, particularly for lean development environment like startups.
Only two personas were used to have a good representation of actual users and evaluate mobile application. Since its difficult for startup to engage real users in the development process for time and economic reason, APs provide an opportunity for startups organization developing early-stage software. It can be an alternative that is fast at execution (avg 1.14 seconds), economic (GPT-4o required a monthly subscription to be used around 20$), and reasonably reliable.
Although they cannot be considerate as a full alternative for human input given their AI-based nature, but they might bring value as a complementary tools alongside real user engagement.
On the other hand, the codebase analysis AP doesn't really apply to startup context. The main reason being their long execution time providing less flexibility and adaptability. The quality of their output needs further depth and analysis to be relevant to software practitioners.
As LLMs continue to be updated, younger open-source models such as Llama, Mistral, and DeepSeek are likely to improve in performance and relevance relative to OpenAI's GPT. Their open-source nature also makes them especially attractive for software startups, both for economic reasons and for maintaining privacy in R&D.

This research project lays out the foundation for experimenting Active persona feedback generation, it can be reproduced by implementing your own persona and provide your own screenshot for analysis (API key required for GPT). Regarding the codebase analysis it can be reproduced by downloading open-source model via Ollama. It applies mostly for android application but can be modified to suit your codebase.

## 5.3 Reflecting On Crafting Effective Prompts

The central role of prompt engineering has been acknowledged during the experimentation phase regarding the performance of Active Personas. Despite the innovation in Gen AI, models like GTP 4-o and LLama3 generated feedback quality still rely heavily on the structure, clarity, and objectiveness of the input prompt. Poor designed prompts lead to vague, generic and misleading claims filled with hallucinations. This is an issue that undermine the value of APs concept, in the following selection the best practices for prompt design are presented and put in contrast against ineffective examples.

### 5.3.1 Best Practices in Prompt Engineering

1. **Clarity and Explicitness**

   - Effective prompt
     "*Analyze the readability of this Bitwarden password display, specifically focusing on whether characters can be easily distinguished. Mention accessibility concerns that could affect user trust."*

   - Ineffective prompt
     "*Analyze Bitwarden password display and tell me what you think*."

The ineffective prompt is a vague and specific enough, the AP will generate a shallow response that will be very generic and prone to hallucination. With such a prompt the AP lack direction and will default to generic feedback that stay too much on the surface and won't be able to detect any issues or provide new insights.

2. **Persona Implementation**

   - Effective prompt
     *"You are Alex, a Gen Z user familiar with Telegram and WhatsApp,you like secure apps and looking for a good messaging app, evaluate whether Signal's interface supports fast onboarding, minimal distractions, and a polished navigation experience."*

   - Ineffective prompt
     "*Give me your thought on the UI, review the components and give me user feedback based on your analysis*."

Implementing a persona greatly implement the quality of the generated feedback and make the feedback more personal and in alignment with the persona values and goals. Without it, there is no alignment with the intended real users target, and the model defaulted to an unengaged tone, missing key usability criteria.

### 3. Case Study

- Effective prompt
  *"Focus on whether special characters such as '@', '<', and '§' are displayed correctly in both view and edit modes of the password field. Note any inconsistencies and their implications for user trust."*
- Ineffective prompt
  "*Check if there is any issues or improvement on the current view."*

The ineffective prompt often failed to surface the actual rendering bug (e.g., the missing '<' character). Instead, the AI would default to shallow responses like "The screen seems minimalistic," or "Looks secure." In most cases, it missed the core issue entirely. When it did succeed to detect the issue, it was inconsistent and not reliable. This randomness is unacceptable in systematic evaluations, where consistency and alignment with objectives are critical.

### 4. Expected Output Format

- Effective prompt
  "*-Focus your answer on the UI elements shown in the file and use the file name as an indicator if helpful.*
  *- Only evaluate what is visible in the screenshot — be specific and precise"*

- Ineffective prompt
  "*Provide your thoughts on the UI and UX*"

The ineffective prompt is not detailed enough and will generate output that will be inconsistent in terms of length and prone to hallucination issues, inventing components that are not present in the provided screenshots.

From the related work section, previous studies have established the relationship between effective prompts and maximizing the effectiveness of generative AI models in user-centered design. Idris et al. [17] highlighted the importance of input structuring in co-design workflows, arguing that prompt clarity and accuracy directly influence the AI's ability to generate meaningful design insights. The best practices identified in this section directly build on Idris et al work. By demonstrating and evaluating how different prompt patterns influence the quality and accuracy of the generated feedback by APs, the findings of our study provide concrete, empirically grounded examples of how prompt engineering affects generative performance in UI/UX contexts.

### 5.3.2 Contribution to Research and Practice

To avoid such shortcomings and obtain reliable output from AP, the four steps introduced should be consistently applied when designing prompts. These include clarity and explicitness, persona implementation, case-specific focus, and defined output format. Applying these guidelines will not only results in reducing vague, hallucinated responses but will ensures feedback aligned with real users and evaluation goal. This contributes to more consistent and actionable outputs, especially valuable in resource-constrained environments like startups.

By providing these guidelines extracted from the learning of the demonstration of the DSR framework, this research provides a practical foundation in integrating APs in the most efficient way.

# 6 CONCLUSION AND FUTURE WORK

## 6.1 Conclusion

The thesis was conducted to explore the potential of Active Persona in generating feedback that are contextual and qualitative to solve an ongoing issue in the cycle of software development – the lack of user involvement in the development process, focusing on startup environment where resources are often constrain and where implementing effective user engagement session is costly and time consuming. By investigating the concept of persona and then Active Persona, Generative AI coupled with the concept of persona seemed like an interesting investigation to solve the issue of user involvement. Throughout the development of the DSR research methodology valuable artifacts were developed to simulate user interactions and generate contextual feedback that match with established guidelines *ISO 9241-112:2017(E) and ISO 9241-110:2020* but also representative of real user feedback.

The primary research question (**RQ1**) asked to what degree AI-powered Active Personas can generate contextual user feedback that aligns with real user needs and contributes to product quality. The AP that implemented Persona A and B (representing actual users of the app targeted) provided qualitative and contextual feedback that aligned well with real user needs. Based on visual cues (screenshots of the app) the app was able to use computer vision to detect UI components effectively. The Active Persona often discerned the same issues as real users and go beyond their feedback by providing solid recommendations for improvement, recommendations on what modifications should be adopted for better usability and interaction. Those recommendations can be considered valuable when it comes to improving the quality of the software product. The quality of the generated feedback for the most part feel very human and required not a lot of data to be effective. And the execution time for the APs analysis is very impressive and provide flexibility, aspects that might appeal to startup organization regarding the optimization of their workflow. The analysis of the generated feedback for the open-source models highlighted some critical nuance to the effectiveness of APs. While the feedback generated via visual images and interaction-based evaluation are surprisingly contextual, well presented, and coherent, the codebase focused evaluations from APs were often too generic, speculative feedback without actionable recommendations, which leads us to research question 2.

The secondary research question (**RQ2**), which addressed the systematic limitations and areas for improvement in AI-driven user interaction simulation. The primary issue is the genericity of the codebase analysis feedback from APs. The evaluation failed to accurately trace UI-related issues back to the corresponding sections of code. The feedback analyzed are generic guidelines that are speculative and does not provide an evaluation that can be trusted. This issue was mostly tested on open-source models that are new to the market of LLMs, their capacity for interpreting and reasoning through complex application logic is limited and time consuming. This approach to AP evaluations is inefficient for a startup, leading to wasted time rather than productivity gains. Time and improvement on the LMMs are required to consider the codebase analysis approach reliable and trustworthy with the output. Another limitation to the quality of the feedback regrading user interaction simulation is the design of the prompt itself. Prompt engineering played a crucial role in obtaining qualitative feedback by framing AI reasoning and attention on specific UI elements. Poorly designed prompt during the investigation resulted in inaccurate and often incomplete results.

To this day the AI is not fully autonomous enough to detect a specific issue on an app interface without the assistance of a human, emphasizing the continuous need for human supervision in the evaluation process and the involvement of user in the product development feedback loop.

## 6.2 Future Work

When it comes to Active Persona, future work involves making the artifact developed available to the public, most specifically for practitioners that are operating in startup environment, so they will be able to test and adapt the Active Persona to their own context. By creating an HTML website, practitioners will only need to import their own screenshots and select characteristics of their targeted users. After processing the inputs, the website will return the contextual feedback. Further investigation on the codebase analysis is also worth considering especially when interacting with open-source models.

Another direction for pushing this work further will be to consider taking the concept of Active Persona on a bigger scope, now that we can affirm that it is possible for APs to generate contextual and relevant feedback to some extent, a case worth investigating would be embedding Active Personas directly into the software development process loop as a live component. Instead of iteratively send prompts guiding which case to investigate, the AP will be able to run continuously and monitor user flows, providing real-time feedback to practitioners. This approach could be highly effective in shaping more accurate user profiles and collecting both quantitative and qualitative data about user behavior—potentially making APs an integral part of continuous usability testing.

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# 8 APPENDIX

## 8.1 More detailed evaluation from APs

### 8.1.1 Case 1 APs feedback Persona B

**Detailed Screenshot Reviews**

Screenshot: Password profile of John Doe - View.png The screenshot displays the Bitwarden app with a focus on the password section, which reads "sweden0Oskane." In evaluating the readability of the password font:

1. **Font Distinguishability**:

– The lowercase "o" and the numeric "0" have sufficient differentiation. The "0" is slightly taller and oval, whereas the "o" is more rounded. This characteristic reduces potential confusion.

– The uppercase "O" is also distinguishable from both "0" and "o," appearing more oval and larger in comparison to the lowercase letter.

2. **Color and Clarity**:

– The black font color against a white background provides good contrast, aiding in readability.

– The font size appears reasonable for most users, but users like me, who prefer larger text, might use accessibility settings to increase it. Even at this size, there's clarity.

3. **Potential Confusion**:

– For people with vision impairments or those unfamiliar with typing differences, there's a moderate risk of mixing these characters up, particularly if font size or contrast settings are reduced from their current state.

4. **Trust and Readability**:

– The font feels professional and clean, not giving off any mistrust. Its simplicity aids in deciphering characters accurately without unnecessary strain.

In summary, the password font in the Bitwarden app is generally clear and distinguishable for crucial characters, addressing primary concerns of readability and trust. Potential risks exist primarily when font size is decreased or users do not have experience distinguishing typographical differences. It might be beneficial to provide an option for enhanced character distinction in settings for older or visually impaired users.

Screenshot: Password profile of Lin Nux - View.png In evaluating the password font readability on the Bitwarden app, several factors stand out based on the provided screenshot.

Font Type and Character Distinction:

The password "1LvwvSB8D1Lcvbfeu" uses a sans-serif font, which generally aids in clarity. However, there are specific concerns with distinguishing visually similar characters such as:

- "1" (one) vs. "l" (lowercase L): The delineation is reasonably clear, with the "1" having a slight serif and the "l" being more rounded and simple. However, for aging users or those needing enhanced readability, the similarity could cause confusion.
- "0" (zero) vs. "O" (uppercase o): This particular password does not show these characters, but ensuring distinct shapes and widths when present usually helps.
- "I" (uppercase i) vs. "l" (lowercase L): This isn't illustrated directly, but distinguishing via height or slight serifs can prevent misinterpretation.

Readability and Trust:

The text color is black on a white background, providing high contrast and aiding readability. The consistent font size appears adequate for typical use, although for someone regularly using increased font sizes, the option to adjust this within the app would enhance accessibility. Visibility is further supported by decent spacing between characters, minimizing potential misreads.

Potential Flaws:

While the font choice mostly supports distinguishing similar characters, reliance on slight nuances like mild serifs might not be sufficient for all users, especially those with visual

impairments. Consideration for a more pronounced differentiation (e.g., slashed zeros or unique glyphs for certain letters) would enhance usability.

**In conclusion, while the current design is mostly user-friendly, it could benefit from refined clarity in character distinction to support users who prioritize readability and simplicity. Addressing these issues holistically would enhance trust and ease of use for a demographic that includes users like Geraldine.**

### 8.1.2 Case 5 APs detailed feedback Persona B

**Detailed Screenshot Reviews**

Screenshot: 1-vault- main view of the app.png This screenshot shows the Bitwarden app interface, focusing on the "My Vault" section.

**Modern, Intuitive, and Accessible Elements**:

1. **Layout and Design**: The interface is clean and uncluttered, with ample white space, making it visually appealing and easy to read. The use of simple icons and clear labels enhances intuitiveness.
2. **Navigation**: The bottom navigation bar offers clear and consistent access to key functions, a modern design choice that helps users easily switch between tasks.
3. **Font and Icons**: The text size appears adequate, and icons are simple and recognizable, important for users who rely on accessibility features like increased font size.

Outdated or Hindering Components:

1. **Visibility of Actions**: While the three-dot menus (ellipsis) next to entries suggest additional actions, their purpose isn't immediately clear at a glance, potentially hindering quick task completion.
2. **Color Contrast**: The color contrast between text and background seems appropriate, although ensuring optimal contrast for all elements (like the ellipsis) is vital for accessibility.

Overall, the Bitwarden app's UI design appears modern and functional, but slight adjustments to improve the visibility of certain interactive elements could enhance usability, especially for users who prioritize simplicity and clarity.

Screenshot: 2-Send View.png The Bitwarden app screenshot showcases a clean and modern interface with clear typography and simple organization. The UI elements, such as buttons and icons, appear intuitive and easily identifiable. The use of a prominent blue header contrasts well with the white background, enhancing readability and visual hierarchy.

The bottom navigation bar is straightforward, featuring recognizable icons with labels like "Send" and "Settings," which aligns well with current design trends prioritizing clarity and ease of access. The "+" button for adding new items is distinctly marked and easily accessible.

However, there might be some room for improvement regarding space utilization. The lists and icons could be more spaced apart to aid users, particularly those who rely on increased font sizes for accessibility. The menu dots on the right could potentially be larger for better touch interaction.

Overall, the app's interface looks intuitive and user-friendly, minimizing technical jargon and unnecessary complexity, catering well to the demographic that values simplicity and straightforward navigation.

Screenshot: 3-setting view.png The Bitwarden app UI seen in the screenshot reflects a clean and intuitive design, which aligns well with contemporary user needs for simplicity and accessibility. The use of a straightforward list format for settings ensures clarity and ease of navigation. Each setting category is clearly labeled, which minimizes the cognitive load on the user.

Visually, the design is modern due to its minimalistic approach, using ample white space to enhance readability, especially for users who may increase font size via accessibility settings.

The color scheme is simple, with a strong contrast between the blue header and white background, aiding in both focus and readability. Icons in the bottom navigation bar are easily recognizable and clearly labeled, which helps users quickly understand their function.

However, while the design is functional and user-friendly, there is a lack of graphical elements or animations that might enhance engagement without complicating usability. Overall, the design effectively meets the essential requirements for accessibility and ease of use, particularly important for users like you, who value simplicity and consistency.

Screenshot: 4-new send view.png The Bitwarden “New Send” UI screenshot demonstrates a clean and straightforward design, which aligns well with the needs of users seeking simplicity and ease of use.

**Modern and Intuitive Elements:** - Color Scheme and Typography: The use of a blue header bar with white text is visually appealing and aids readability. The clear distinction between sections and labels helps in easy navigation and comprehension. - Buttons and Toggles: The toggle switch for “hide the text by default” is intuitive and follows modern design conventions. The selectable buttons for “File” and “Text” types are clearly defined and easy to interact with.

**Accessible Features:** - Clear Labels and Instructions: Labels such as “Name,” “Type,” and “Text” are straightforward, accompanied by brief explanatory texts, minimizing confusion for users unfamiliar with technical jargon. - Spacing and Layout: Adequate spacing between elements ensures the UI is not cluttered, enhancing focus and making it easier for users to identify different sections without being overwhelmed.

**Potential Improvement Areas:** - Font Size: While generally clear, offering larger font options could enhance readability for users who turn on accessibility features frequently. - Interactive Elements Visibility: The “Options” dropdown could be more visually prominent to signal interactivity, possibly using an icon or color accent to indicate it clearly as an interactive element.

Overall, the UI elements appear modern and aligned with current user needs, supporting an accessible and user-friendly experience without overwhelming the user with complexity.

Screenshot: 5-view - generator.png The Bitwarden app’s UI shown in the screenshot demonstrates a clean and functional design, which caters well to users like Geraldine who value simplicity and clarity. The app employs a modern approach with a minimalistic design, ensuring a focus on usability and intuitive interaction. The elements are well-spaced, with a clear hierarchy and a distinct color scheme that aids in differentiating controls and text.

The use of toggle switches for options like “A-Z,” “a-z,” and others is consistent with contemporary design trends, providing a straightforward means of enabling or disabling features. The slider for password length is intuitive, allowing users to easily adjust the settings.

The iconography at the bottom navigation bar is simple and recognizable, facilitating easy navigation between different sections like “My vault,” “Send,” “Generator,” and “Settings.” However, the presence of some technical terms, like special characters, might slightly challenge users who are less tech-savvy. Overall, the design is not cluttered with unnecessary elements, thereby minimizing cognitive load.

One potential area for improvement could be providing explanatory tooltips or help icons for each feature to aid users who might not be familiar with password requirements or options. In summary, the UI is modern and well-aligned with current user needs, maintaining accessibility and an intuitive flow.

### 8.1.3 Case 6 APs detailed feedback Persona A

Screenshot: 1- main view.png The Signal app's UI in this screenshot presents several strengths and a few areas for potential improvement, fitting well within modern design standards. The layout is clean, with ample spacing, contributing to an intuitive and uncluttered experience. This minimalistic approach helps in focusing on essential tasks, which is appreciated by users who value efficiency.

Key UI elements such as the search icon and the three-dot menu are recognizable, providing easy access to additional features without overwhelming the screen. The use of familiar icons for functions like chats, calls, and stories at the bottom further enhances navigation. The presence of profile icons adds a personal touch while maintaining clear user identification.

However, there are slight improvements that could be considered. The "Get started" section at the bottom, while offering clear onboarding options, uses buttons that could possess a more contemporary style in terms of color and shape. The current look appears slightly outdated and might not align with the slickness observed in the rest of the app. Additionally, the camera and edit buttons floating within the chat list are functionally sound but could be more embedded into the overall design to appear less intrusive.

**Overall, the UI is modern and visually appealing, efficiently balancing functionality with aesthetics. Refining a few components could further enhance the seamless and attractive experience that tech-savvy users like me expect.**

Screenshot: 2- calls view.png The Signal app's UI in this screenshot has a clean and modern feel, which aligns well with contemporary design trends. The use of white space provides a clear and uncluttered look, enhancing readability and user focus. The navigation bar at the bottom divides sections into Chats, Calls, and Stories, using simple, intuitive icons paired with text labels for clarity.

The layout is straightforward, with elements like the search icon and menu neatly positioned for easy accessibility. The call button is prominently placed, suggesting a user-centric design aimed at facilitating quick actions. The use of subtle color accents on icons helps distinguish different functions without being overwhelming.

However, there are some areas for improvement. The color contrast could be more pronounced to enhance visibility, especially for users with visual impairments. The icons and text appear somewhat small, which might affect usability on devices with smaller screens.

**Overall, the design is visually appealing and functional, though slight adjustments in size and contrast could improve accessibility further. There are no apparent outdated design patterns, and the interface aligns with modern UI standards, supporting swift interaction for tech-savvy users like myself.**

Screenshot: 3-grroupe info view.png The UI of the Signal app here looks clean and modern, reflecting contemporary design trends. The use of soft, rounded buttons for "Story," "Video," "Mute," and "Search" is visually appealing and adds a friendly, approachable feel. The icons are intuitive and clearly convey their functions, supporting quick comprehension.

The layout is well-organized, with clear sections for settings like "Disappearing messages," "Chat color & wallpaper," and "Sounds & notifications." This makes navigation straightforward and helps users quickly find what they need without unnecessary clutter.

The spacing between elements is well-balanced, preventing any feeling of congestion or overcrowding, which enhances readability and usability. The subtle color palette maintains focus on the content without causing distraction.

There's no sign of outdated design patterns; instead, the interface feels efficient and tailored for fast-paced use. The overall simplicity and focus on essential features enhance the user experience, aligning well with the expectations of a tech-savvy user like myself.

**One minor area for improvement could be adding more customization options directly visible on this screen, but overall, the UI seems to meet modern standards effectively.**

Screenshot: 4-setting view.png The UI of the Signal app's settings screen appears clean and modern, aligning with the expectations of someone who values efficiency and clarity. The layout is organized with a clear hierarchy, featuring well-spaced elements that avoid clutter and allow

for straightforward navigation. The use of intuitive icons next to each menu item enhances visual appeal and aids in quick identification of options.

Text and icons are clearly legible, with adequate spacing that promotes accessibility. The minimalist design avoids unnecessary distractions, focusing on core functions efficiently. The color scheme is simple yet effective, providing a professional look that aligns with modern design trends, without diverting attention from essential tasks.

One potential area for improvement could be the inclusion of more dynamic elements or color highlights for increased engagement or easier focus areas, although this simplicity can also be a strength for users who prefer distraction-free interfaces.

**Overall, the Signal settings UI is robust and user-friendly, likely meeting the needs of a tech-savvy individual seeking smooth and intuitive digital experiences**.

Screenshot: 5-story guidance instruction.png The displayed UI of the Signal app presents a modern and intuitive interface. The layout is clean with a minimalistic design, which aligns well with contemporary design standards that emphasize simplicity and ease of use.

UI elements such as the icons for gestures (tap, swipe up, swipe right) are clear and internationally understandable, promoting seamless interaction without needing much explanation. The use of white, easy-to-read text against a blurred background ensures good contrast, enhancing accessibility.

The “Got it” button is aptly positioned for easy acknowledgment without distraction. The action indicators are centrally aligned, which draws immediate attention and provides a straightforward guide to the user’s next steps. This design choice enhances user flow and reduces potential confusion over in-app navigation.

The interface avoids clutter, which is crucial for maintaining focus and efficiency. There are no unnecessary distractions, and the instructions are concise. However, ensuring that the these gestures are familiar to users can further improve accessibility, especially for those less familiar with swipe-based navigation.

**Overall, the UI exhibits a modern design ethos with a strong emphasis on user engagement and comfort. The interface feels up-to-date with current trends and sufficiently intuitive to accommodate the expectations of Gen Z users like myself who value efficiency and sleek design in technology.**

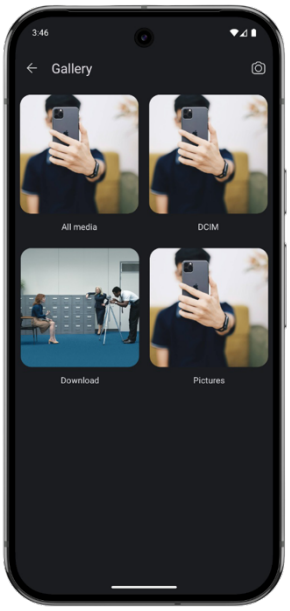
## 8.2 Case 4 Screenshots Provided




Step 0 - Initial conversation view



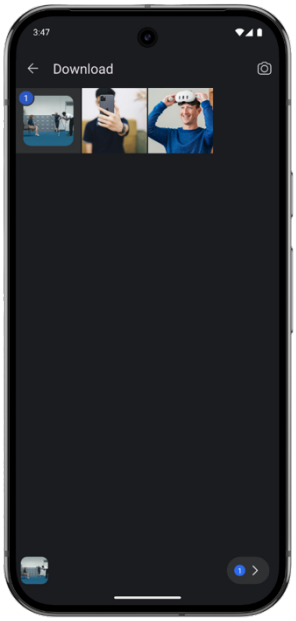
Step 1- Camera App




Step 2 - Gallery Selector



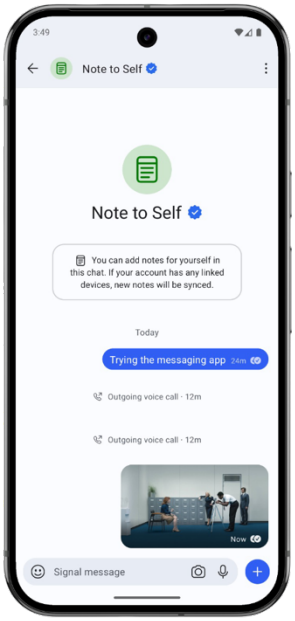

Step 3 – Folder Selection




Step 4 – Photo Selection

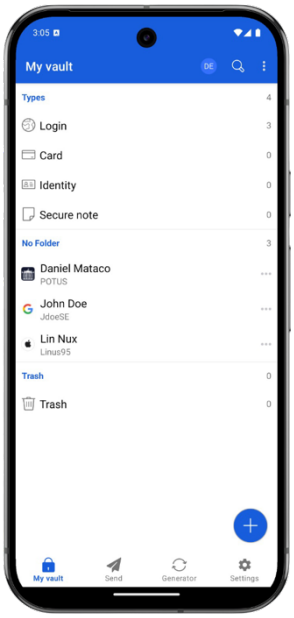



Step 5 – Photo Validation and/or Edition

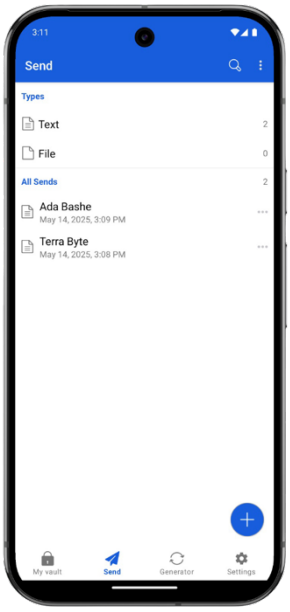



Step 6 – Photo sent back to inital view

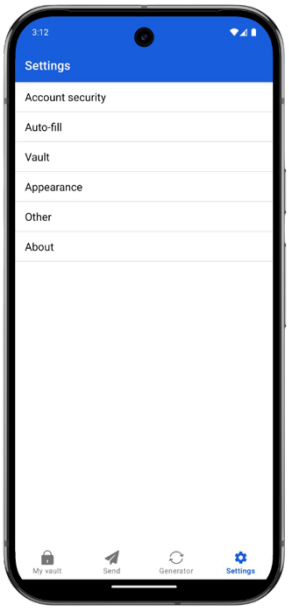
## 8.3 Case 5 Screenshots Provided



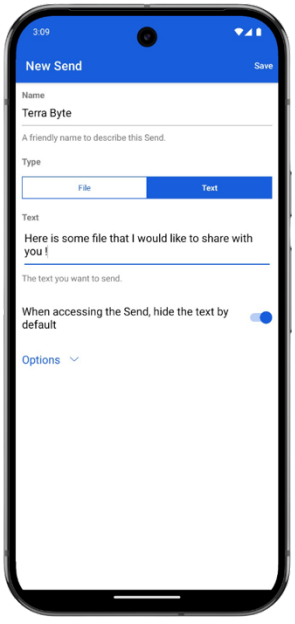

Main View of the app – My vault



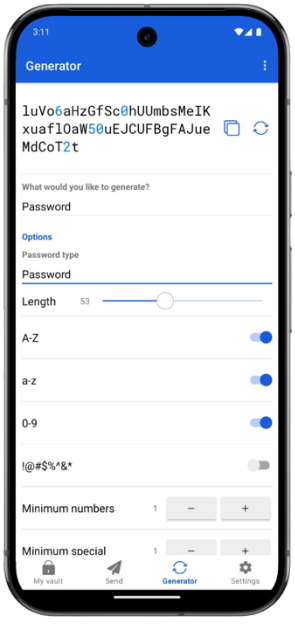

Send View




Settings View

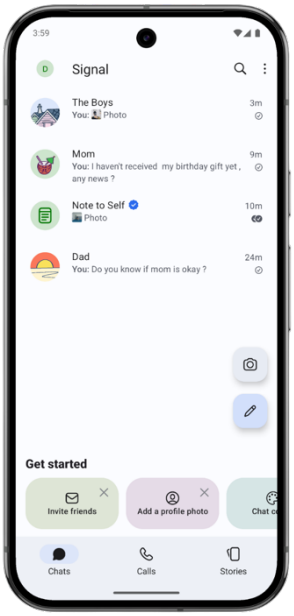



New Send View




Password Generator View

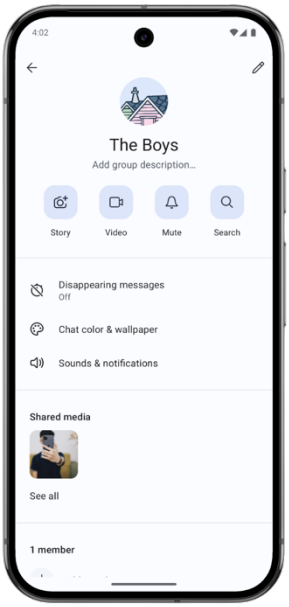
## 8.4 Case 6 Screenshots Provided



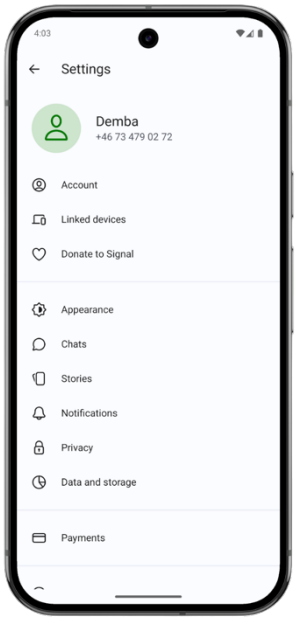

Main View of the app – Chats



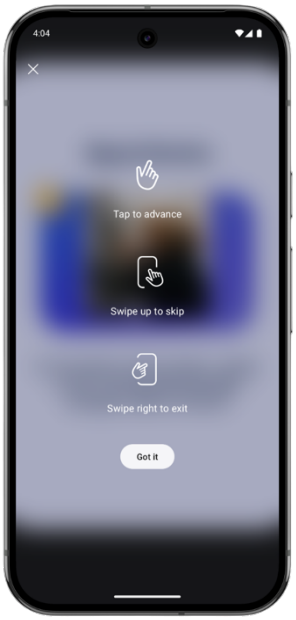

Call View




Group Settings View




Settings View




Tutorial View

## 8.5 Bitwarden PlayStore Reviews

The full reviews file can be found with that link:
https://drive.google.com/file/d/148E1pnRDXxS1KIvF3Q9NB6X4TSZ0kuVd/view

| | |
|---|---|
| 2 | previous issue I had emhas been fixed, great app and browser extension absolutely love it and that it's open source and free!! |
| 3 | Love it. Has totally simplified my online life. |
| 4 | few recent updates after the ui changes has made the app so buggy. it crashes every time when i try to use fingerprint to unlock and gives error while accessing the vault |
| 5 | the best. |
| 6 | No Way to do a Password Recovery, I lost all my passwords. Contact Support did not even respond to my email. |
| 7 | Does exactly what it says it does. Easy to use. |
| 8 | Autofill have stopped working recently in my microsoft adge browser. I have tried turning it on and off again and giving the accessibility setting again but no luck. |
| 9 | Making minimum OS version Android 10 without making a lite version of the app is disappointing. |
| 10 | best app |
| 11 | Simply : 🙂. |
| 12 | Best |
| 13 | I don't know where to leave a bad review for the browser add-on for PCs so I'm leaving this here. The new addon changed all the features that used to take 1 click to 2 or 3 clicks. Change functionality of clicking on entire entry from filling to a tiny little button. Got rid of the shortcuts for copying a username and password and made it a drop-down box. This is idiotic and needs to go back the way it was |
| 14 | Mtls support would make it 5 star. To bad this still isn't working and limits the usage for secure remote use |
| 15 | Bitwarden is great! This app allows you to confidently and securely store your passwords/ID cards. You can set up your phone so it'll automatically search for passwords in Bitwarden when you're trying to log in to a website/app. Bitwarden also has an extension you can use on a PC. Most impressively, you can set up a 2nd form of security when you're trying to log in to websites such as your bank, which will make it more secure. I'd definitely recommend this app, like my dad did for me. |
| 16 | The new UI is near unusable, those responsible should be fired. I have since switched to proton pass. |
| 17 | I don't like the mobile app as much as the proper PC app. Also the PC app has been updated with a "new look". Give me back to old look & the old operation I say. |
| 18 | Bitwarden is an awesome open source password manager. As long as they stay OSS, I will continue to suggest them to people who need a password manager. They also take payment in Bitcoin which is awesome 😎😎 |
| 19 | Been saving my life for the past 5 years. |
| 20 | Broken since the UI update. Still a good app, just doesn't unlock occasionally and the only fix is to seemingly reboot your entire phone. Will change review when its fixed (if I remember lul)... |
| 21 | Amazing |
| 22 | Awesome PW manager. I've used a pay-for competitor for years and haven't regretted the switch to Bitwarden once. |
| 23 | Getting it to work as seamlessly on android or IOS as it does on say a windows laptop but it just doesn't. Otherwise it's a very helpful app. |
| 24 | I thought it was amazing at first but now I'm starting to hate it. It doesn't autofill many websites or apps that it should, it doesn't autosave passwords, often even after manually filling out the password and details the manager erases it after double checking because I didn't press save right away but if I press save right away then I have to presave a password that doesn't yet exist then go hunt it down, copy it and set it to the website password or app password. I'm going to switch. |
| 25 | Awesome password safe and app. For some reason it doesn't consistently offer to auto fill passwords like LastPass used to. |
| 26 | A solid app, but I did find a bug. If you have a password with a less-than symbol (i.e. <) in it, and you click on the eyeball to display the password, the string that it displays will be missing the less-than symbol. If you edit the password, you will see the less-than symbol. Please fix! Thank you! |
| 27 | I've been using Bitwarden for more than a year now, and I love the open source behind it and the community that makes it possible. That being said, I'll switch to another, more reliable pass manager. Too often it happens, with Bitwarden, that autofill doesn't work. It becomes less practical and convenient. For my liking. I'll probably swing back in a year or two, if it's still around. But for now I just find it more often frustrating thanx convenient. |
| 28 | Love the app bcoz of it's features but it often stops responding & has to be stopped. This mostly happens when pincode is used to unlock the app. It shutters each time & if the 'Unlock' button pressed few too many times after pin is entered the app promptly crashes. It also shutters a lot during normal operations as well .e.g. during opening folders or editing login details. |
| 29 | Simply the best password manager, there is no need to buy anything. This is perfect! Works in iOS, android,Mac,Windows, every platform it even has an extension, so you can autopilot easily on Mac and Windows so there's no need to type your passwords in your notes because bitwarden exists! |
| 30 | 希望能有按月收费订阅 应用非常方便，多端同步，非常 nice |
| 31 | Can't live without this app. |
| 32 | The best password manager! |

## 8.6 Signal PlayStore Reviews

The full reviews file can be found with that link:
https://drive.google.com/file/d/1RIwnXL9G015LRiW8T12TslVMnsrMQqzp/view?usp=sharing

| | | |
|---|---|---|
| 2 | 2025-03-29 20:43:46 | Good messaging app that keeps your chats private. As long as you don't have people in your group that take screen shots and post them. |
| 3 | 2025-03-29 20:41:48 | I can't transfer accounts between my machines so I'm very unhappy |
| 4 | 2025-03-29 20:40:19 | Awesome app works real well. |
| 5 | 2025-03-29 20:35:51 | A completely free, very private and secure messaging app that is also open-source! No complaints really, other than the odd bug here and there. Would be nice to see edge-to-edge enforcer status and navigation bars) not solid coloured. |
| 6 | 2025-03-29 20:20:39 | Great app. We use it for all messaging, you should too! |
| 7 | 2025-03-29 20:10:07 | awesome |
| 8 | 2025-03-29 20:02:26 | I Want Share My Screen It's Can't Possible |
| 9 | 2025-03-29 19:39:30 | I do not wish to write a full in depth if you value Security in your life this will help |
| 10 | 2025-03-29 19:35:39 | it messes up my phone by giving me advertisements |
| 11 | 2025-03-29 19:14:46 | fine |
| 12 | 2025-03-29 18:54:06 | Good |
| 13 | 2025-03-29 18:47:47 | So far so good. I've had it for years but dormant. Now it is my main text app (I use audio messages a lot). And a frivolous thing to add, but we love it: you can choose your logo color of the a app that is too common a color for apps is now a light purple on my phone screen. Easier to locate. |
| 14 | 2025-03-29 18:40:54 | How do I get into a private.... Nevermind 🤔 |
| 15 | 2025-03-29 18:39:29 | super |
| 16 | 2025-03-29 18:11:43 | very new to this app. so far, it's great. |
| 17 | 2025-03-29 17:55:16 | very good |
| 18 | 2025-03-29 17:36:19 | I have been scammed on here |
| 19 | 2025-03-29 17:31:23 | I use it for sending memes, but turns out I'm enjoying the same level of security as the highest military officials in the world. |
| 20 | 2025-03-29 17:02:43 | awesome 👌 |
| 21 | 2025-03-29 16:57:32 | Audio and video quality is not good as compared to other applications |
| 22 | 2025-03-29 16:41:15 | Always working. No print screen possible! |
| 23 | 2025-03-29 16:17:42 | It nice and I love it using it |
| 24 | 2025-03-29 16:16:55 | Marvelous |
| 25 | 2025-03-29 15:50:54 | smooth and easy |
| 26 | 2025-03-29 15:35:22 | awesome |
| 27 | 2025-03-29 15:01:37 | voice are not good |
| 28 | 2025-03-29 14:55:37 | privacy ❤️❤️ |
| 29 | 2025-03-29 14:52:38 | Great app for messaging family, sharing military secrets, war plans, espionage, etc. The worst part is accidentally adding a reporter to your not so secret chat, on purpose. But, its like accide to your grandmother. We've all done that. At the end of the day you just find a little bit about yourself. Like how you and your friends are morons, or that your member matches your grandfat You make everything Great Again! - no nudes were actually sent to family. |
| 30 | 2025-03-29 14:31:03 | super , no doubt about 👌 security 🔐. |

## 8.7 Active Persona Artifacts

The implementation for the Active Persona and screenshots used can be found with that link: https://github.com/DembaSE/Active-persona.git